

Next Generation Data Center Operations for SAP Enterprise Software

Module 01

Virtualization, Linux & SAP HANA



Curriculum version

- Curriculum Version:
 - Short Version - 2604
- Software used:
 - MobaXterm (Windows)
 - Terminal App (Mac)
- Prerequisites
 - [Downloaded eduVPN](#)



Authors

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- Dr. Holger Wittges (TUM)



Target audience

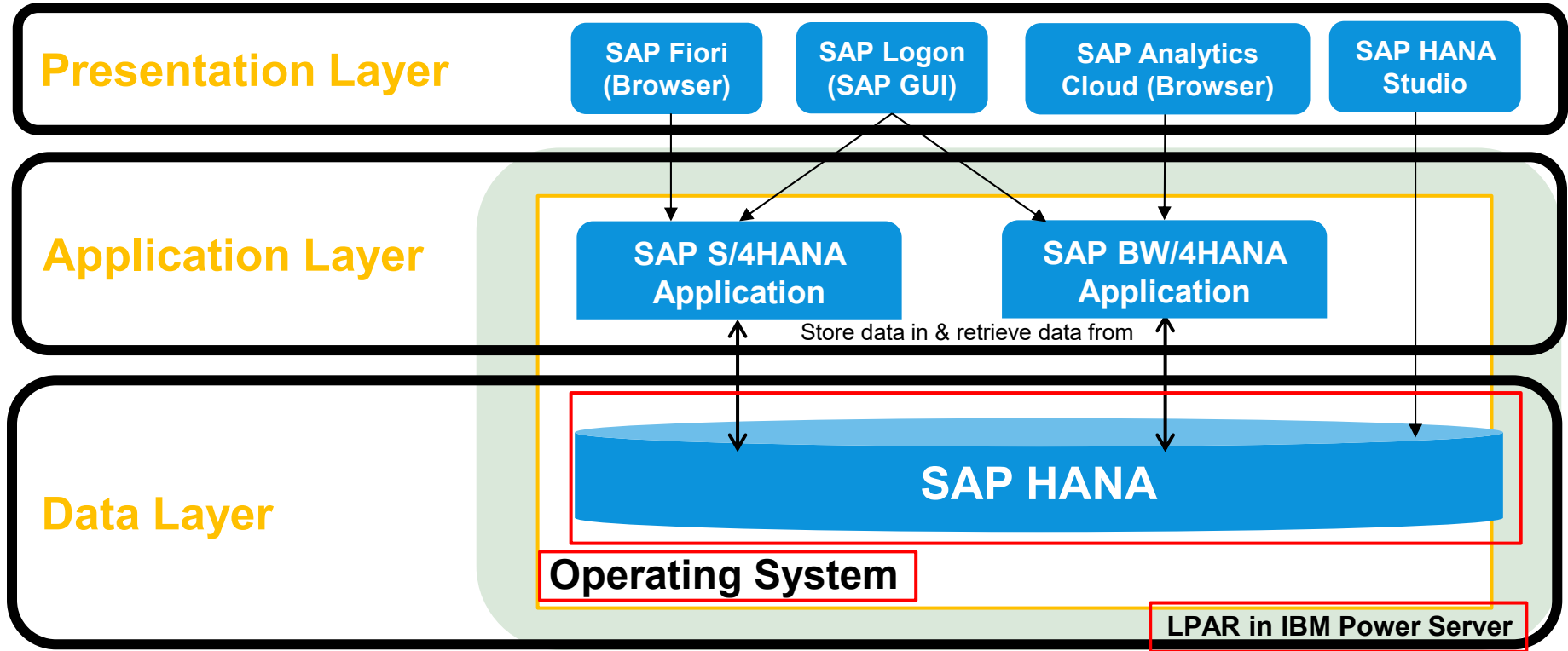
- SAP Business Users and Lecturers
- Information Systems (IS) Students
- Computer Science (CS) Students
- Management Students



Learning objectives

- Instead of a theory-heavy, frontal-teaching of SAP HANA, we will make you familiar with the technical processes that are usually hidden behind the surface of SAP systems.

Introduction

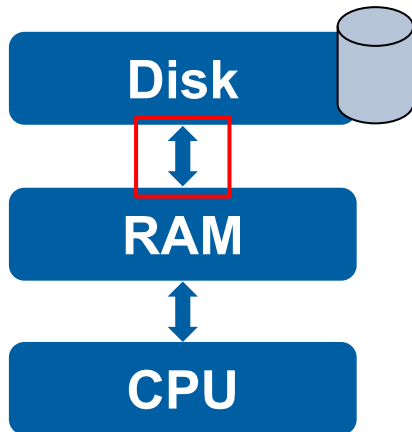


„Technical Foundations“

Image Source: Self-created

Introduction

- Without high-capacity RAM
 - Database stored on disk
 - **Bottleneck**: Latency between disk and RAM



- With high-capacity RAM (SAP HANA)
 - Data stored in-memory
 - Bottleneck: Latency between CPU and RAM
 - Huge response time improvement

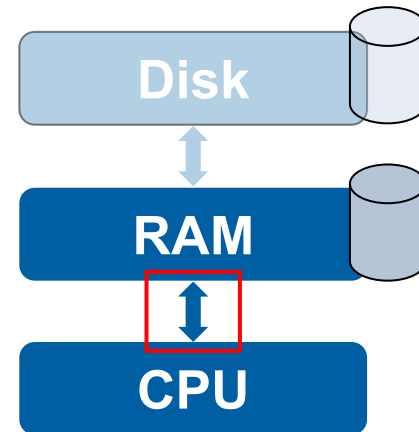


Image Source: Self-created

■ SAP HANA Cloud

- Hosted by hyperscaler, such as Microsoft, Amazon, etc.
- You don't have to worry about the operating system level and the infrastructure.
- **Challenges:**
 - Data Privacy
 - Latency based on your internet connection
 - Availability is not “in your hands”

■ SAP HANA On-Premise

- Self-hosted, e.g. on IBM Power servers:
 - Full control
 - Specifically Power servers have many other advantages, e.g.:
 - **Reliability** of 99,999% (~5 minutes downtime per year)
 - **Flexibility**: With a single keystroke, IBM Power servers can change the allocated processor capacities of instances (virtual machines) with a granularity of 0.01 cores.
 - You need employees with **administration skills** (this curriculum is just a first step)

Research Group (SAP UCC) Server Room:
We also host our SAP services on-premise using IBM Power.

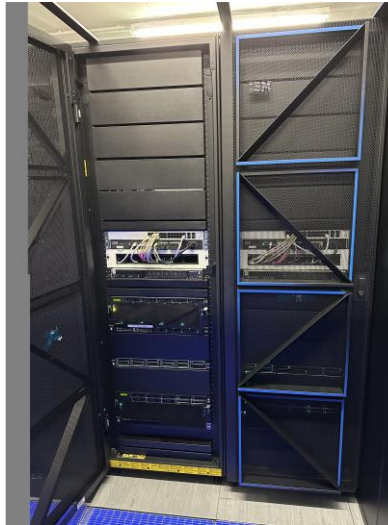
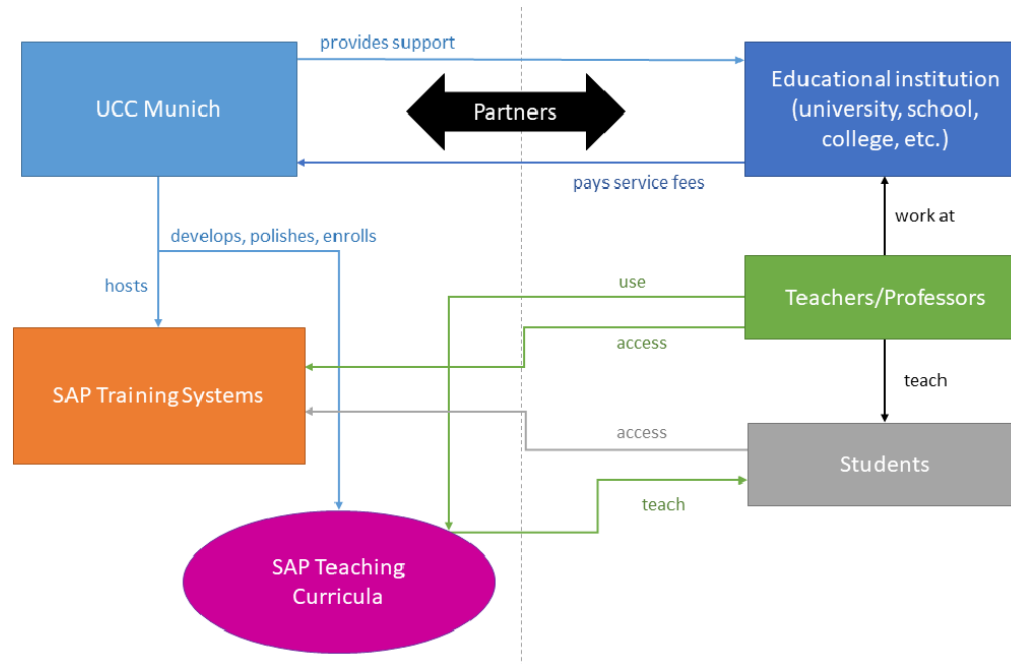


Image Source: Self-created

Introduction



Hardware



„the UCC serves as an Education-as-a-Service provider for SAP products“

Sources:

https://de.m.wikipedia.org/wiki/Datei:IBM_logo.svg

& [SAP S/4HANA on Red Hat OpenShift Container Platform: Business Perspective from a Cloud Hosting Provider](#)

Introduction

- **2x IBM Power E1050 server, per unit:**
 - CPU (Installed / Max): **72 / 96 Cores**
 - Memory (Installed / Max): **8 / 16 TB**
 - Memory bandwidth: 3.276 GBps
 - IO bandwidth: 756 GBps
- **2x IBM Storage Flashsystem 5200, per unit:**
 - Maximum bandwidth (reads): 21 GBps
 - Response times (reads): < 50 μ s
 - Capacity (Installed / Max):
138 TiB (~100 TB) / 340 TB



Sources: [IBM Power E1050: Technical Overview and Introduction](#)
& [IBM Storage FlashSystem 5200 Product Guide](#)

Agenda

- **IBM Power Virtualization**
- Fundamentals of Red Hat Enterprise Linux (Hands-On!)
- Podman Containers
- SAP HANA Deployment (Hands-On!)
- Next Time: IBM Granite / LLM (Hands-On!)

Curriculum Structure

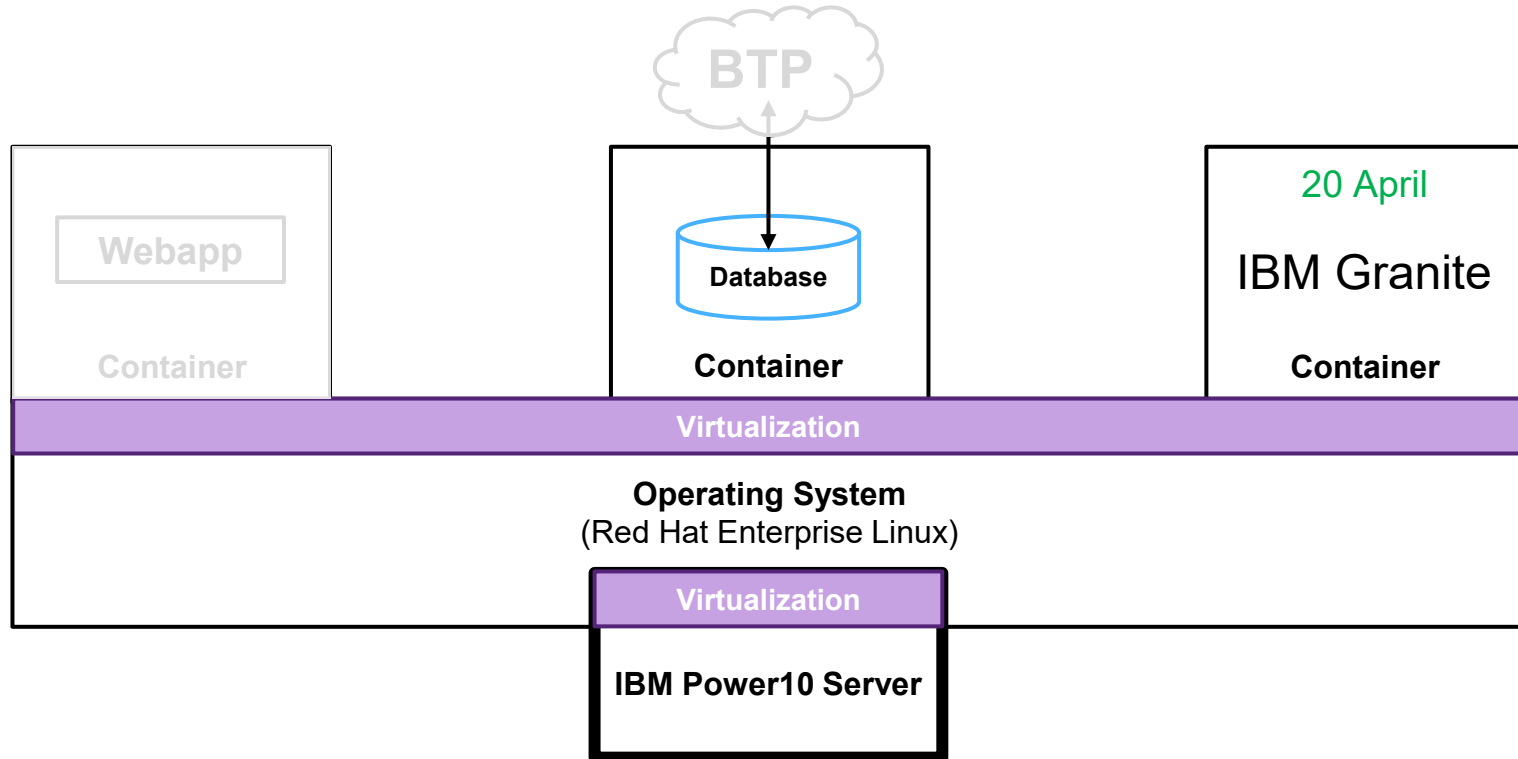
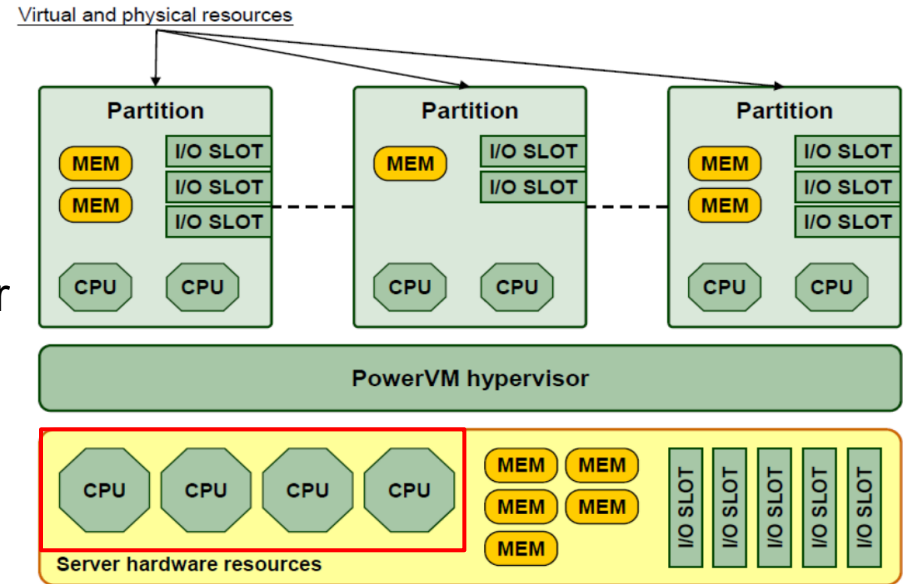


Image Source: Self-created

PowerVM: Overview

- The PowerVM Hypervisor divides physical system resources into isolated Logical Partitions (LPARs).
- Each LPAR operates like an independent server that runs its own operating system, such as Red Hat or SUSE.
- The Hypervisor can assign dedicated processors, memory, and I/O resources, which can be dynamically reconfigured as needed to each LPAR.



Processor Virtualization

Image Sources:

[Introduction to IBM PowerVM](#)

https://de.wikipedia.org/wiki/Datei:Red_Hat_logo.svg

PowerVM: Processor Virtualization

- Dedicated Processors
 - Whole processors are assigned to a single logical partition (LPAR).
 - When assigning dedicated processors to an LPAR, you must assign processors in whole numbers with at least one processor.
 - Likewise, removing processors from a dedicated LPAR, you must remove processors in whole numbers starting with one processor.

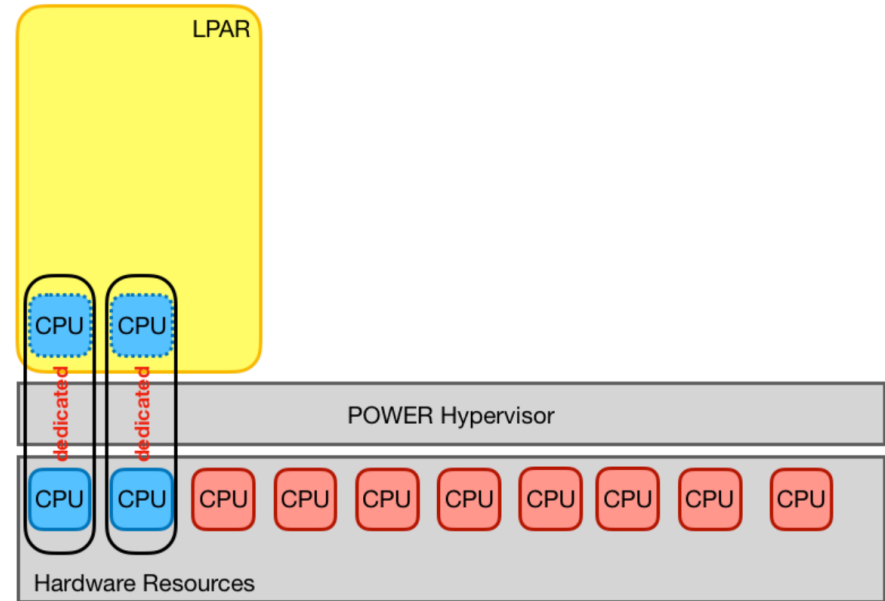


Image Source: <https://powercampus.de/en/documentation/virtualization-with-powervm/practical-introduction/dedicated-processors/>

- Dedicated Donating Processors
 - Dedicated donating refers to the ability to **donate idle or spare processors** that are owned by a dedicated LPAR to a Shared Processor Pool (SPP). The LPAR **maintains absolute priority** over the processors that it owns. The donated processors are returned instantaneously to the dedicated-donating LPAR when it needs them back.
 - Enabling this feature can help to increase global system utilization.
 - Example: ([Link](#))
 - Up to **16 productive SAP HANA LPARs** in dedicated-donating mode can be operated on just one POWER9 server. Additionally, unlimited productive small HANA instances can run via SPP.
 - Compared to the x86 architecture, POWER servers show faster processing and response times as well as a **performance gain of up to 30 percent**.

PowerVM: Processor Virtualization

- Dedicated Donating Processors
 - When restarting the SAP HANA database, we see a peak in the usage of processing units (3,650).
 - During normal operation, the LPAR only requires 1,5 – 2 units and remaining units can be donated.

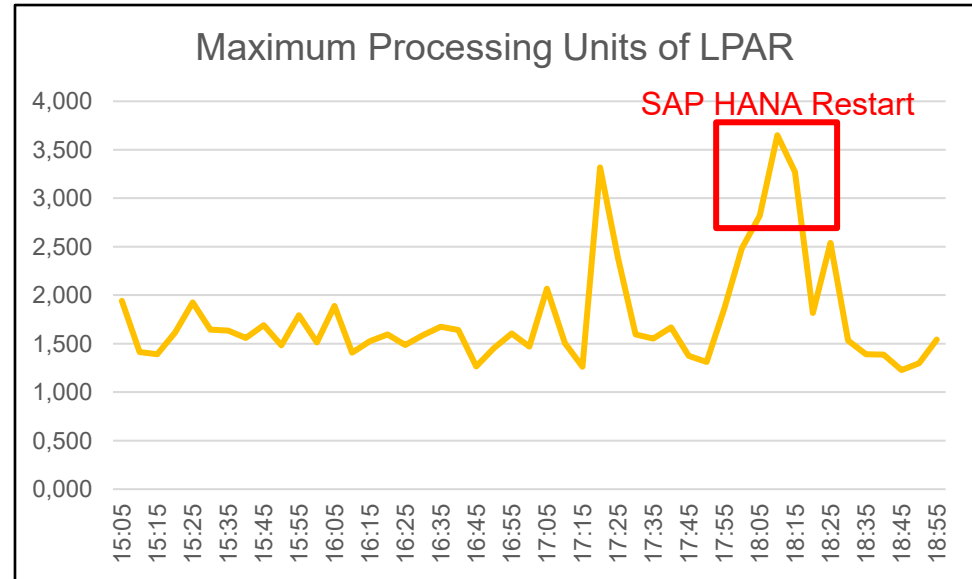


Image Source: Self-created

PowerVM: Processor Virtualization

- Shared Processors
 - Physical processors whose processing capacity is shared among multiple LPARs. The ability to divide physical processors and share them among multiple LPARs is known as Micro-Partitioning.
 - Shared processors can either be **capped** or **uncapped**.

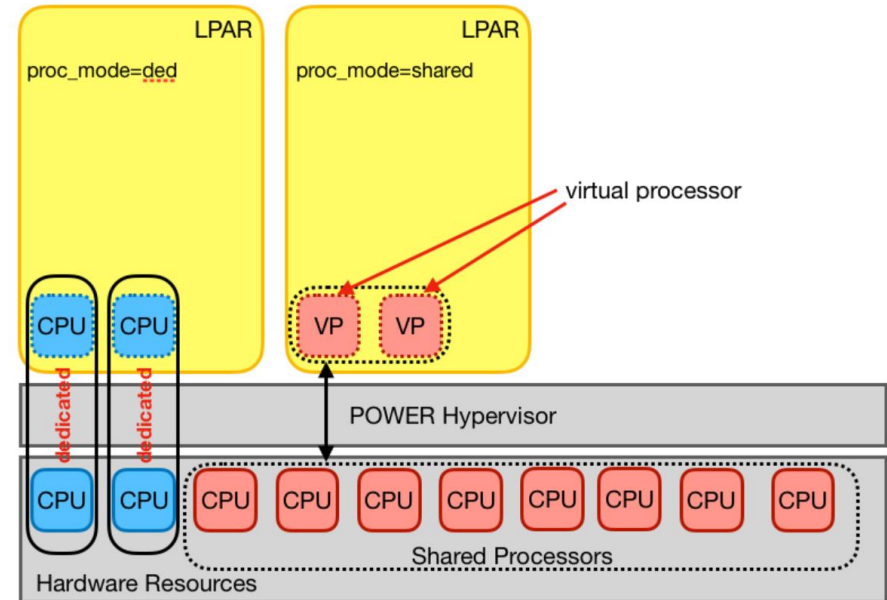


Image Source: <https://powercampus.de/en/documentation/virtualization-with-powervm/practical-introduction/shared-processors-micro-partitioning>

PowerVM: Processor Virtualization

- Capped Shared Processors
 - A capped partition cannot use more than its entitled capacity regardless of how much idle capacity there is in the SPP.
 - Remaining CPU resources are not allocated to a capped LPAR.
 - Guaranteed CPU resources that are not used by a capped LPAR are returned to the SPP and can be used by LPARs with uncapped sharing mode.

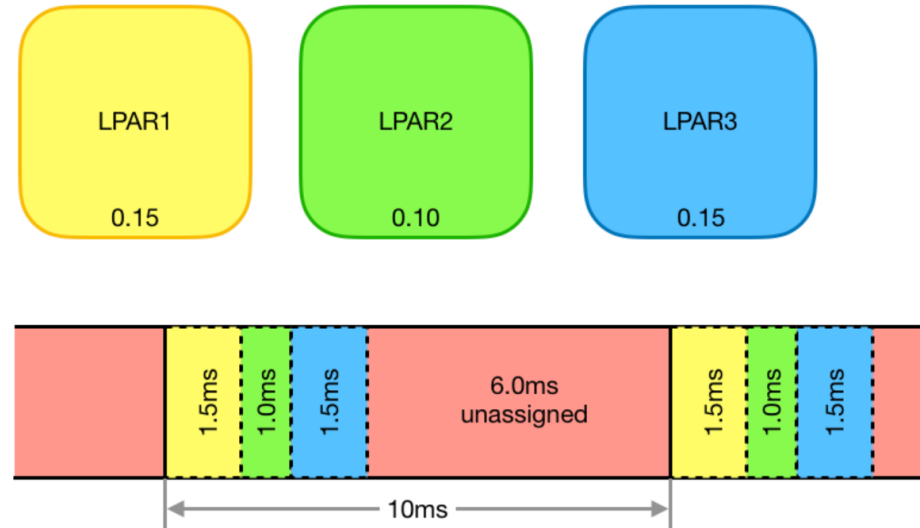


Image Source: <https://powercampus.de/en/documentation/virtualization-with-powervm/practical-introduction/processor-sharing-mode/>

PowerVM: Processor Virtualization

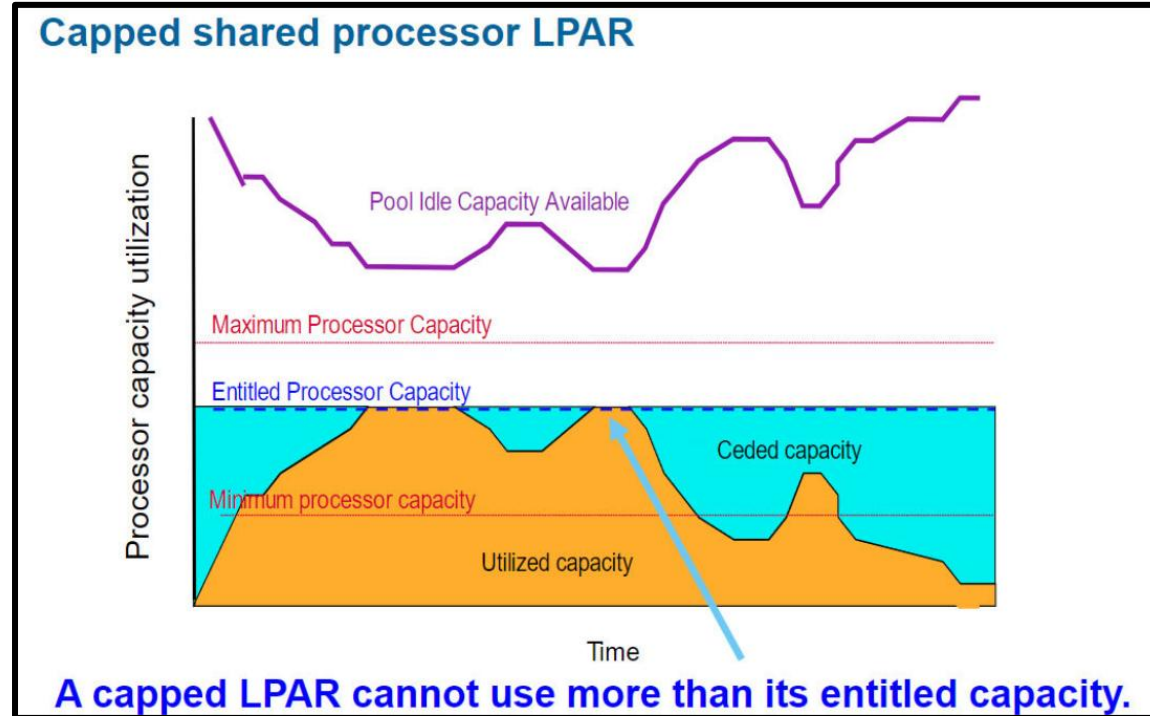


Image Source: [Introduction to IBM PowerVM](#)

- Uncapped Shared Processors
 - In addition to the guaranteed processor performance, LPARs with uncapped sharing mode compete for the remaining CPU time.
 - How much of the remaining CPU time is distributed depends on three factors:
 - Amount of additional CPU time available that is not allocated to an LPAR.
 - The weighting of the LPAR. It can be set to a value between 0 and 255 for each LPAR with shared processors without capping using the `uncapped_weight` attribute (the default value is 128). The higher the value, the higher the share that the LPAR receives.
 - How many other LPARs are currently competing for additional CPU time and their weighting. An LPAR that is idle does not compete for additional CPU time.

PowerVM: Processor Virtualization

■ Uncapped Shared Processors

- The additional CPU time is determined as follows:
 - The weights of all LPARs competing for additional time are added.
 - The share of an individual LPAR results from the quotient of its own weight and the sum of the weights of all competing LPARs.
 - The share is multiplied by the additional CPU time available.
- Example:
 - Sum weights: $150 + 50 + 100 = 300$
 - Shares (UW / 300):
 $1/2$ (LPAR1), $1/6$ (LPAR2), $1/3$ (LPAR3)
 - Times (Multiply by 6ms):
3ms (LPAR1), 1ms (LPAR2), 2ms (LPAR3)

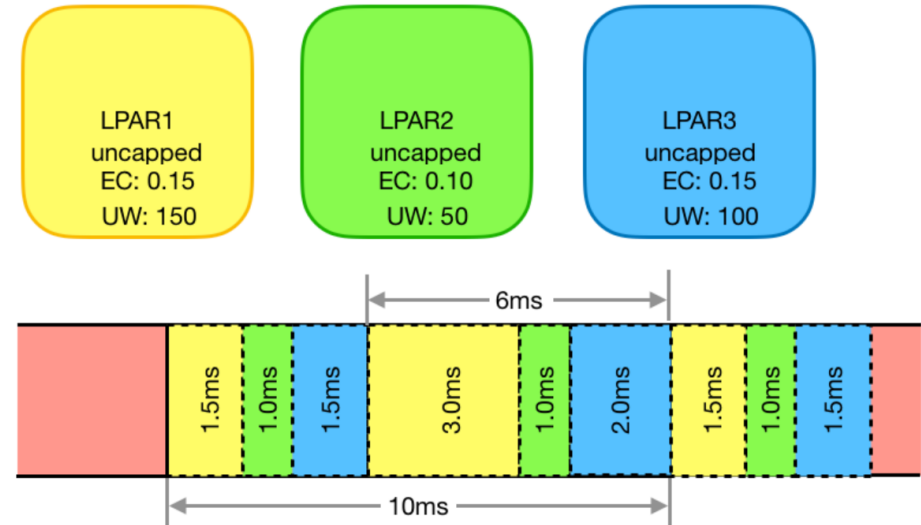


Image Source: https://powercampus.de/en/documentation/virtualization-with-powervm/practical-introduction/sharing-mode-uncap-and-uncapped_weight

PowerVM: Processor Virtualization

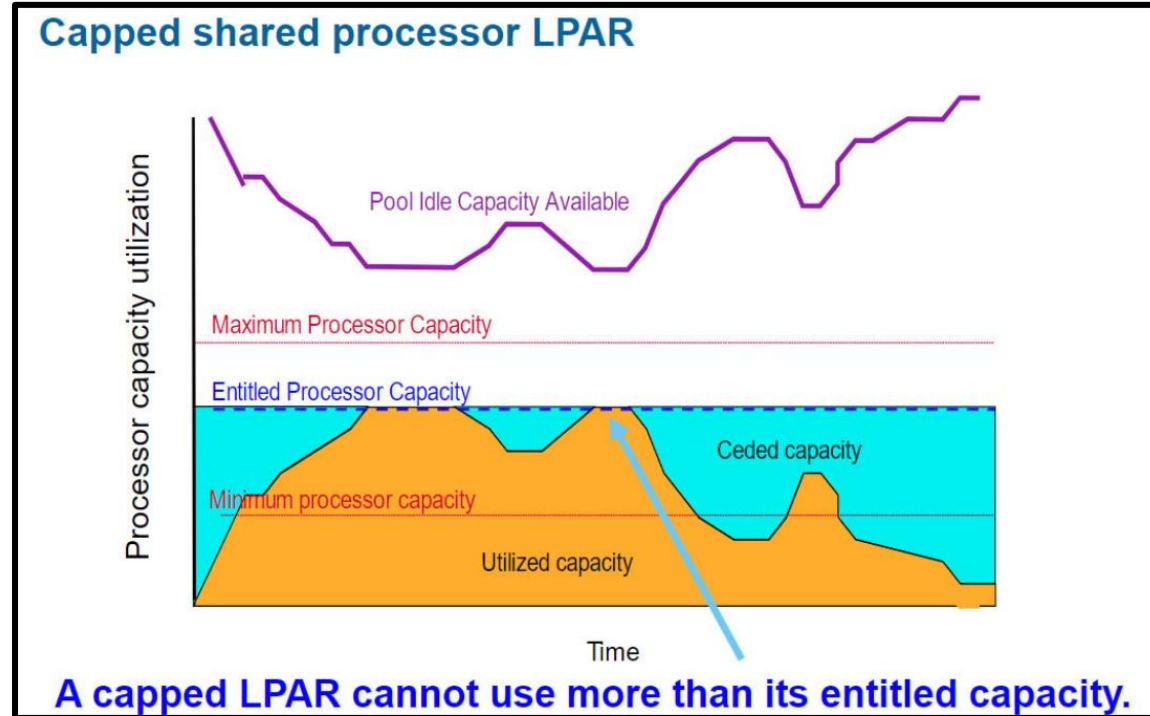


Image Source: [Introduction to IBM PowerVM](#)

PowerVM: Processor Virtualization

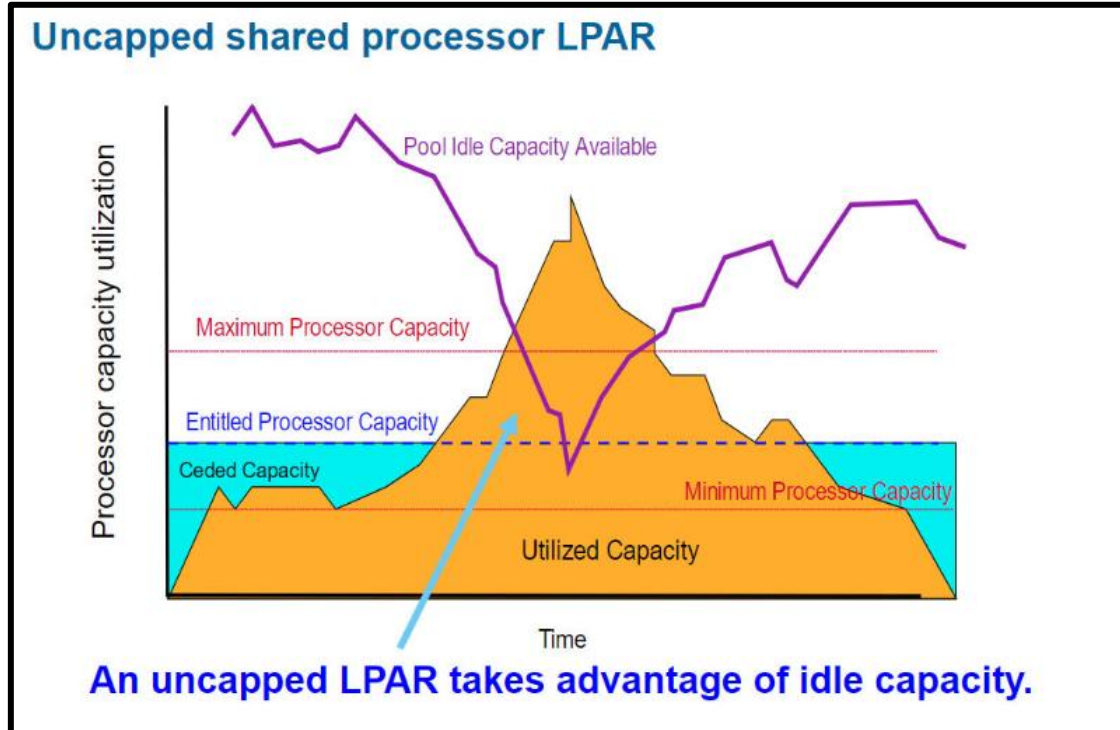


Image Source: [Introduction to IBM PowerVM](#)

- Processor Virtualization Summary:
 - **Dedicated Processors:** Whole processors are assigned to single LPAR.
 - **Dedicated Donating Processors:** Donate spare processors to Shared Processor Pool (SPP).
 - **Shared Processors:** Divide physical processors and share them among multiple LPARs.
 - Capped Shared: LPAR can not use more than its entitled capacity.
 - Uncapped Shared: In addition to the guaranteed processor performance, LPARs with uncapped sharing mode compete for the remaining CPU time within the SPP.

PowerVM: Processor Virtualization

■ Hands-On Exercise:

- The following table shows CPU allocations to 3 different LPARs:

LPARs	Time	1	2	3	4	5	6	7	8
	TEST	2.00	1.70	1.80	2.00	2.00	2.00	1.50	1.40
	PROD	4	5	5	5	4	3	5	5
	DEV	2.50	1.50	1.20	0.20	2.80	3.50	1.20	1.60

- How many CPU cores are in use?
- Give an example which processor types the LPARs could have?
- What is the minimum size of the total shared processor pool (SPP) in CPU cores?

PowerVM: Overview

- The PowerVM Hypervisor divides physical system resources into isolated Logical Partitions (LPARs).
- Each LPAR operates like an independent server that runs its own operating system, such as Red Hat or SUSE.
- The Hypervisor can assign dedicated processors, memory, and I/O resources, which can be dynamically reconfigured as needed to each LPAR.

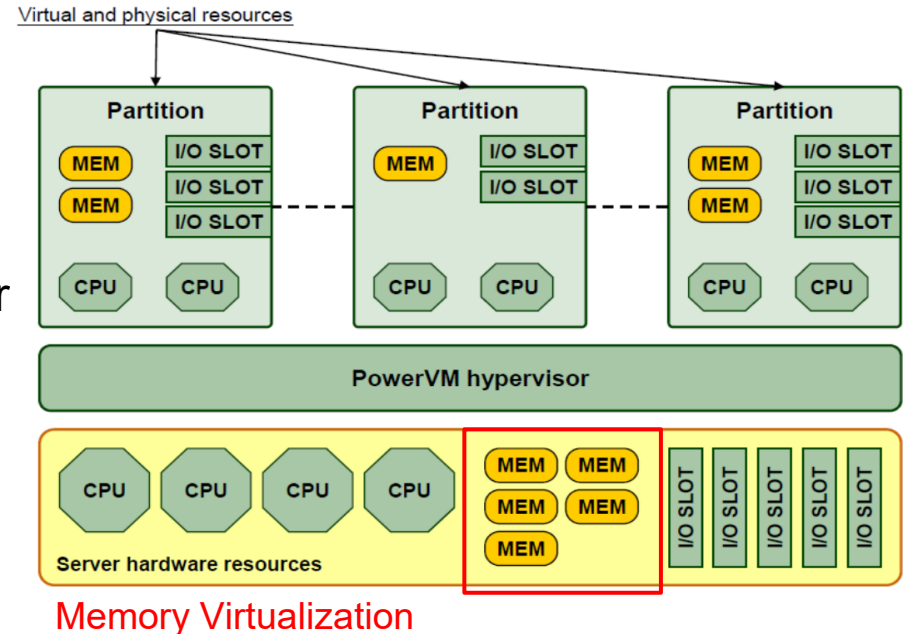


Image Sources:

[Introduction to IBM PowerVM](#)

https://de.wikipedia.org/wiki/Datei:Red_Hat_logo.svg

PowerVM: Memory Virtualization

- „Dedicated“
- Virtual Persistent Memory (vPMEM)
 - Data stored in Virtual Persistent Memory remains persistent as long as the PowerVM hypervisor stays operational.
 - Memory data is NOT erased by a restart of the application accessing them, nor by a Linux OS (i.e. LPAR) reboot.
 - SAP HANA is an in-memory database, which usually needs to reload all data from a disk when restarting.
 - Virtual Persistent Memory enables **SAP HANA Fast Restart**.

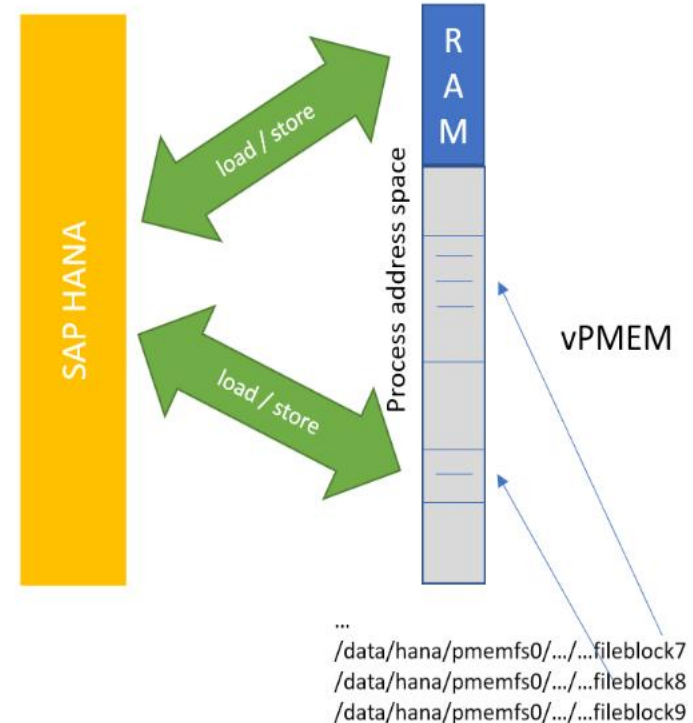


Image Source: Virtual PMEM and SAP HANA Primer

PowerVM: Overview

- Storage Virtualization
- Network Virtualization

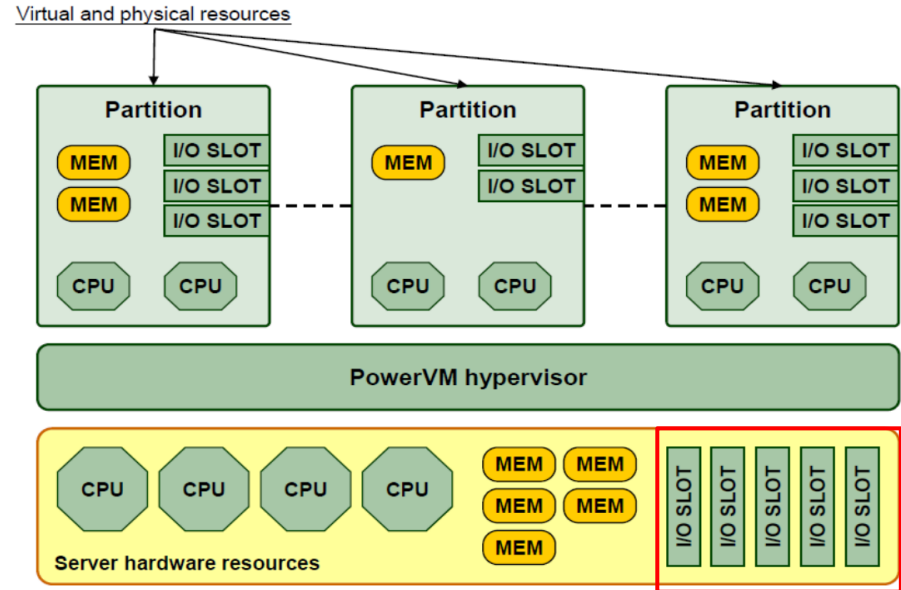


Image Source:
[Introduction to IBM PowerVM](#)

PowerVM: Overview

- Each of you will have access to the same very large partition that we have already provided for you.
- Each participant will install their **own, small, containerized SAP HANA Express** database system on this partition.
- First, we will cover the topics:
 - Linux
 - Containerization

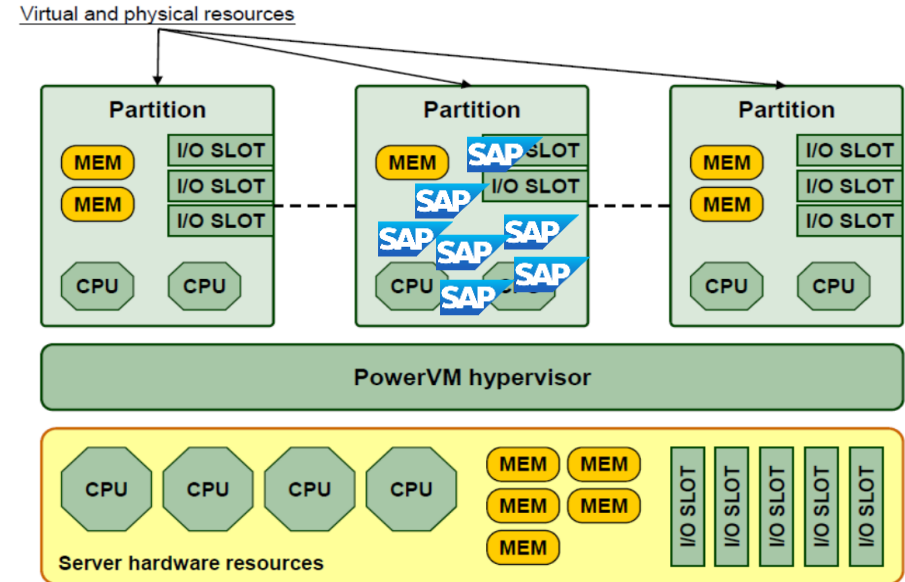


Image Sources:

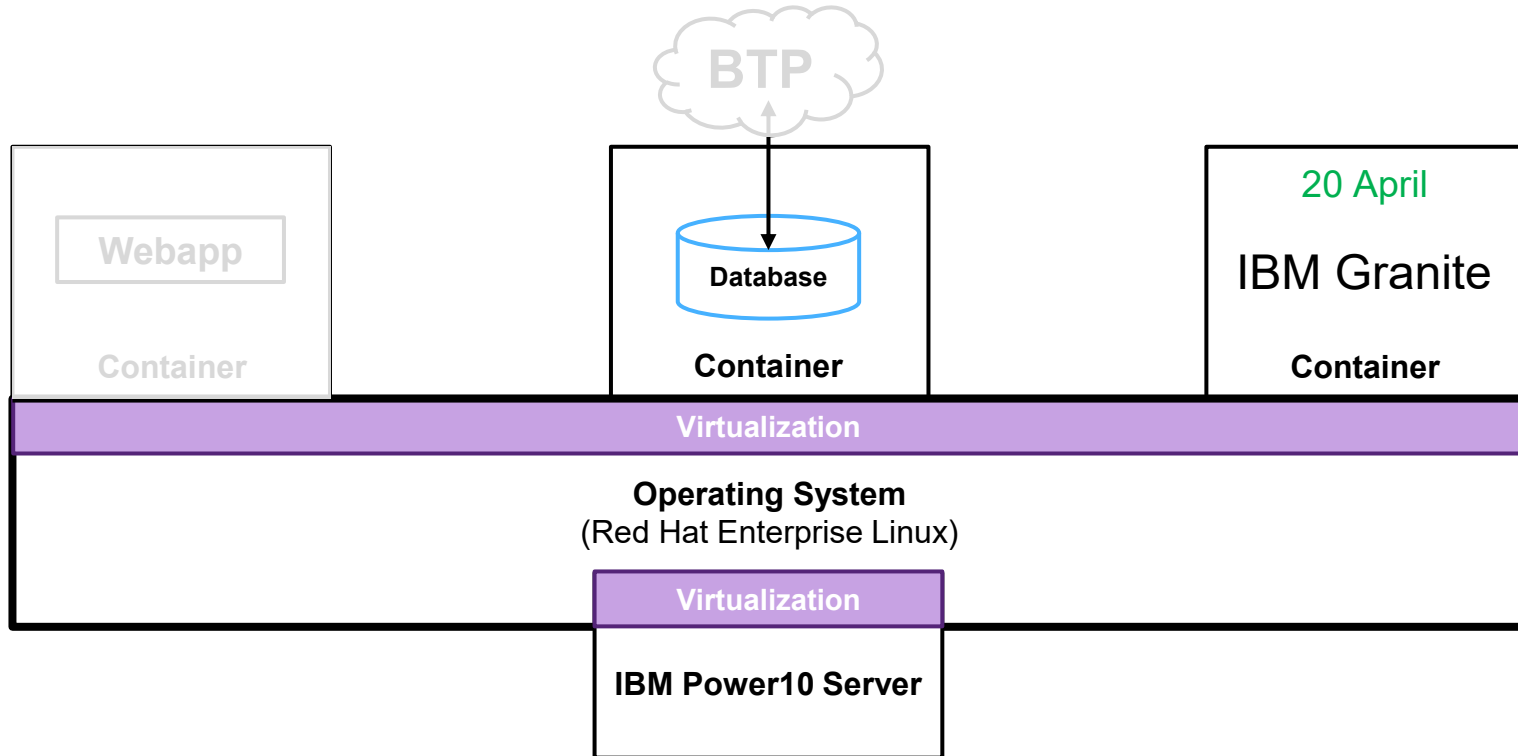
[Introduction to IBM PowerVM](#)

https://de.wikipedia.org/wiki/SAP#/media/Datei:SAP_2011_logo.svg

Agenda

- IBM Power Virtualization
- **Fundamentals of Red Hat Enterprise Linux (Hands-On!)**
- Podman Containers
- SAP HANA Deployment (Hands-On!)
- Next Time: IBM Granite / LLM (Hands-On!)

Curriculum Structure



Linux Distributions for SAP Deployment

- Linux distributions for SAP system administration:
 - RedHat and SUSE both offer an OS officially certified for SAP deployment:
 - *RedHat Enterprise Linux for SAP Solutions* ([Docs](#))
 - *SUSE Linux Enterprise Server for SAP Applications* ([Docs](#))



Image Sources:

https://upload.wikimedia.org/wikipedia/commons/4/44/SUSE_logo.svg

https://de.wikipedia.org/wiki/Red_Hat_Linux#/media/Datei:Red_Hat_logo.svg

Secure Shell (SSH)

- What we will be doing next:
 - As already said, we provided an LPAR on our IBM Power server for you.
 - In the next steps, you will connect to the LPAR using SSH.
 - You will all connect to the same LPAR, using different OS user names (see info sheet).

- Make sure that you have installed [EduVPN](#) and that it is currently running.

Terms of Use

- Users must not:
 - Engage in activities that disrupt system performance and availability.
 - **Attempt unauthorized access to other systems** or data.
 - Execute malicious code, hacking attempts, or security breaches.
 - Use the system for personal, commercial, or illegal activities.

- Intentional misuse or violation of these terms may result in:
 - Immediate revocation of access rights.
 - Reporting of the incident to the respective academic institution.
 - Potential legal consequences in cases of severe misconduct.

Terms of Use

- We may use the monitored data to evaluate our teaching process, and may publish the results.
- However, the OS names are anonymized and we will not publish any personal information about you.
- **We encourage you to participate**, and if you are not doing well, that is fine.
Your inputs will not be graded. 😊
- **By accessing the LPAR, you automatically agreed to all Terms of Use.**

Info Sheet

- **Student**

- Number: 03703057

- **General**

- Hostname: **sose26dc.ucc.cit.tum.de**
- OS User: **learn1**
- Initial User Password: **sose26init1**

- **Information for SAP HANA Module:**

- SAP System ID: **HXE**
- Instance Number: **01**
- System Database Name: **SYSTEMDB**
- System Database Port (SYSDB-Port): **30113**
- Tenant Database Name: **HXE**
- Tenant Database Port (TENDB-Port): **30115**
- Database User: **SYSTEM**
- Database User Password (Recommended): **Password1**

Info Sheet

- **Student**

- Number: 03819691

- **General**

- Hostname: **sose26dc.ucc.cit.tum.de**
- OS User: **learn2**
- Initial User Password: **sose26init1**

- **Information for SAP HANA Module:**

- SAP System ID: **HXE**
- Instance Number: **02**
- System Database Name: **SYSTEMDB**
- System Database Port (SYSDB-Port): **30213**
- Tenant Database Name: **HXE**
- Tenant Database Port (TENDB-Port): **30215**
- Database User: **SYSTEM**
- Database User Password (Recommended): **Password1**

Info Sheet

- **Student**

- Number: 03769763

- **General**

- Hostname: **sose26dc.ucc.cit.tum.de**
- OS User: **learn3**
- Initial User Password: **sose26init1**

- **Information for SAP HANA Module:**

- SAP System ID: **HXE**
- Instance Number: **03**
- System Database Name: **SYSTEMDB**
- System Database Port (SYSDB-Port): **30313**
- Tenant Database Name: **HXE**
- Tenant Database Port (TENDB-Port): **30315**
- Database User: **SYSTEM**
- Database User Password (Recommended): **Password1**

Info Sheet

- **Student**

- Number: 03759690

- **General**

- Hostname: **sose26dc.ucc.cit.tum.de**
- OS User: **learn4**
- Initial User Password: **sose26init1**

- **Information for SAP HANA Module:**

- SAP System ID: **HXE**
- Instance Number: **04**
- System Database Name: **SYSTEMDB**
- System Database Port (SYSDB-Port): **30413**
- Tenant Database Name: **HXE**
- Tenant Database Port (TENDB-Port): **30415**
- Database User: **SYSTEM**
- Database User Password (Recommended): **Password1**

Info Sheet

- **Student**

- Number: 03764758

- **General**

- Hostname: **sose26dc.ucc.cit.tum.de**
- OS User: **learn5**
- Initial User Password: **sose26init1**

- **Information for SAP HANA Module:**

- SAP System ID: **HXE**
- Instance Number: **05**
- System Database Name: **SYSTEMDB**
- System Database Port (SYSDB-Port): **30513**
- Tenant Database Name: **HXE**
- Tenant Database Port (TENDB-Port): **30515**
- Database User: **SYSTEM**
- Database User Password (Recommended): **Password1**

Info Sheet

- **Student**

- Number: 03772566

- **General**

- Hostname: **sose26dc.ucc.cit.tum.de**
- OS User: **learn6**
- Initial User Password: **sose26init1**

- **Information for SAP HANA Module:**

- SAP System ID: **HXE**
- Instance Number: **06**
- System Database Name: **SYSTEMDB**
- System Database Port (SYSDB-Port): **30613**
- Tenant Database Name: **HXE**
- Tenant Database Port (TENDB-Port): **30615**
- Database User: **SYSTEM**
- Database User Password (Recommended): **Password1**

Info Sheet

- **Student**

- Number: 03759778

- **General**

- Hostname: **sose26dc.ucc.cit.tum.de**
- OS User: **learn7**
- Initial User Password: **sose26init1**

- **Information for SAP HANA Module:**

- SAP System ID: **HXE**
- Instance Number: **07**
- System Database Name: **SYSTEMDB**
- System Database Port (SYSDB-Port): **30713**
- Tenant Database Name: **HXE**
- Tenant Database Port (TENDB-Port): **30715**
- Database User: **SYSTEM**
- Database User Password (Recommended): **Password1**

Info Sheet

- **Student**

- Number: 03811579

- **General**

- Hostname: **sose26dc.ucc.cit.tum.de**
- OS User: **learn8**
- Initial User Password: **sose26init1**

- **Information for SAP HANA Module:**

- SAP System ID: **HXE**
- Instance Number: **08**
- System Database Name: **SYSTEMDB**
- System Database Port (SYSDB-Port): **30813**
- Tenant Database Name: **HXE**
- Tenant Database Port (TENDB-Port): **30815**
- Database User: **SYSTEM**
- Database User Password (Recommended): **Password1**

Info Sheet

- **Student**

- Number: 03757821

- **General**

- Hostname: **sose26dc.ucc.cit.tum.de**
- OS User: **learn9**
- Initial User Password: **sose26init1**

- **Information for SAP HANA Module:**

- SAP System ID: **HXE**
- Instance Number: **09**
- System Database Name: **SYSTEMDB**
- System Database Port (SYSDB-Port): **30913**
- Tenant Database Name: **HXE**
- Tenant Database Port (TENDB-Port): **30915**
- Database User: **SYSTEM**
- Database User Password (Recommended): **Password1**

Info Sheet

- **Student**

- Number: 03813277

- **General**

- Hostname: **sose26dc.ucc.cit.tum.de**
- OS User: **learn10**
- Initial User Password: **sose26init1**

- **Information for SAP HANA Module:**

- SAP System ID: **HXE**
- Instance Number: **10**
- System Database Name: **SYSTEMDB**
- System Database Port (SYSDB-Port): **31013**
- Tenant Database Name: **HXE**
- Tenant Database Port (TENDB-Port): **31015**
- Database User: **SYSTEM**
- Database User Password (Recommended): **Password1**

Info Sheet

- **Student**

- Number: 03739752

- **General**

- Hostname: **sose26dc.ucc.cit.tum.de**
- OS User: **learn11**
- Initial User Password: **sose26init1**

- **Information for SAP HANA Module:**

- SAP System ID: **HXE**
- Instance Number: **11**
- System Database Name: **SYSTEMDB**
- System Database Port (SYSDB-Port): **31113**
- Tenant Database Name: **HXE**
- Tenant Database Port (TENDB-Port): **31115**
- Database User: **SYSTEM**
- Database User Password (Recommended): **Password1**

Info Sheet

- **Student**

- Number: 03783141

- **General**

- Hostname: **sose26dc.ucc.cit.tum.de**
- OS User: **learn12**
- Initial User Password: **sose26init1**

- **Information for SAP HANA Module:**

- SAP System ID: **HXE**
- Instance Number: **12**
- System Database Name: **SYSTEMDB**
- System Database Port (SYSDB-Port): **31213**
- Tenant Database Name: **HXE**
- Tenant Database Port (TENDB-Port): **31215**
- Database User: **SYSTEM**
- Database User Password (Recommended): **Password1**

Info Sheet

- **Student**

- Number: 03743825

- **General**

- Hostname: **sose26dc.ucc.cit.tum.de**
- OS User: **learn13**
- Initial User Password: **sose26init1**

- **Information for SAP HANA Module:**

- SAP System ID: **HXE**
- Instance Number: **13**
- System Database Name: **SYSTEMDB**
- System Database Port (SYSDB-Port): **31313**
- Tenant Database Name: **HXE**
- Tenant Database Port (TENDB-Port): **31315**
- Database User: **SYSTEM**
- Database User Password (Recommended): **Password1**

Info Sheet

- **Student**

- Number: 03785998

- **General**

- Hostname: **sose26dc.ucc.cit.tum.de**
- OS User: **learn14**
- Initial User Password: **sose26init1**

- **Information for SAP HANA Module:**

- SAP System ID: **HXE**
- Instance Number: **14**
- System Database Name: **SYSTEMDB**
- System Database Port (SYSDB-Port): **31413**
- Tenant Database Name: **HXE**
- Tenant Database Port (TENDB-Port): **31415**
- Database User: **SYSTEM**
- Database User Password (Recommended): **Password1**

Info Sheet

- **Student**

- Number: 03707804

- **General**

- Hostname: **sose26dc.ucc.cit.tum.de**
- OS User: **learn15**
- Initial User Password: **sose26init1**

- **Information for SAP HANA Module:**

- SAP System ID: **HXE**
- Instance Number: **15**
- System Database Name: **SYSTEMDB**
- System Database Port (SYSDB-Port): **31513**
- Tenant Database Name: **HXE**
- Tenant Database Port (TENDB-Port): **31515**
- Database User: **SYSTEM**
- Database User Password (Recommended): **Password1**

Info Sheet

- **Student**

- Number: 03818989

- **General**

- Hostname: **sose26dc.ucc.cit.tum.de**
- OS User: **learn16**
- Initial User Password: **sose26init1**

- **Information for SAP HANA Module:**

- SAP System ID: **HXE**
- Instance Number: **16**
- System Database Name: **SYSTEMDB**
- System Database Port (SYSDB-Port): **31613**
- Tenant Database Name: **HXE**
- Tenant Database Port (TENDB-Port): **31615**
- Database User: **SYSTEM**
- Database User Password (Recommended): **Password1**

Info Sheet

- **Student**

- Number: 03759595

- **General**

- Hostname: **sose26dc.ucc.cit.tum.de**
- OS User: **learn17**
- Initial User Password: **sose26init1**

- **Information for SAP HANA Module:**

- SAP System ID: **HXE**
- Instance Number: **17**
- System Database Name: **SYSTEMDB**
- System Database Port (SYSDB-Port): **31713**
- Tenant Database Name: **HXE**
- Tenant Database Port (TENDB-Port): **31715**
- Database User: **SYSTEM**
- Database User Password (Recommended): **Password1**

Info Sheet

- **Student**

- Number: 03769237

- **General**

- Hostname: **sose26dc.ucc.cit.tum.de**
- OS User: **learn18**
- Initial User Password: **sose26init1**

- **Information for SAP HANA Module:**

- SAP System ID: **HXE**
- Instance Number: **18**
- System Database Name: **SYSTEMDB**
- System Database Port (SYSDB-Port): **31813**
- Tenant Database Name: **HXE**
- Tenant Database Port (TENDB-Port): **31815**
- Database User: **SYSTEM**
- Database User Password (Recommended): **Password1**

Secure Shell (SSH)

- SAP GUI/SAP Fiori: Accessing the SAP application, even if not connected to the same network
 - What if you want to access the underlying Linux system to check logs, manage the SAP kernel or start/stop SAP instances?
- SSH: cryptographic protocol that enables remote connection to other machines
 - MacOS, Windows 10 & later natively support a **ssh** command
 - For older Windows versions, you would need to download and open PuTTY → <https://www.putty.org/> or MobaXTerm → <https://mobaxterm.mobatek.net/>
(If using any Windows version, we recommend to use MobaXTerm)

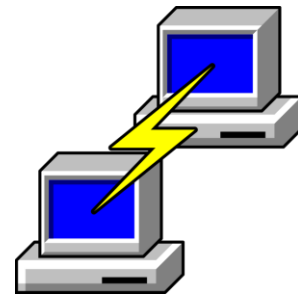


Image Source:
https://commons.wikimedia.org/wiki/File:PuTTY_Icon_upstream.svg

Connecting to the Linux LPAR (Win)

- Open MobaXTerm and set a master password.
 - You can choose this password yourself, but you **need to remember it!**
- Go to *Session* → *SSH*
 - Remote host: *<HOSTNAME>*
 - Check-mark the *Specify username* option
 - Username: *<YOUR LEARN-ID>*
 - Port 22
- Click on *OK*

Connecting to the Linux LPAR (Win)

- For example, if your Learn-ID is learn99 and the hostname is workshop.ucc.cit.tum.de:

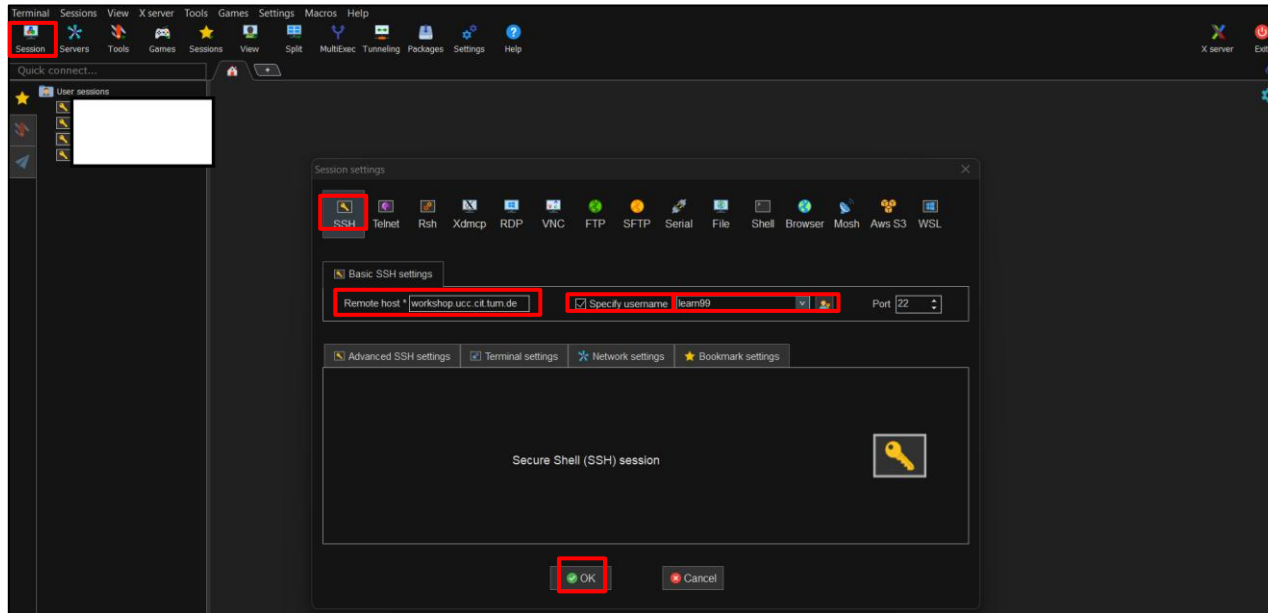
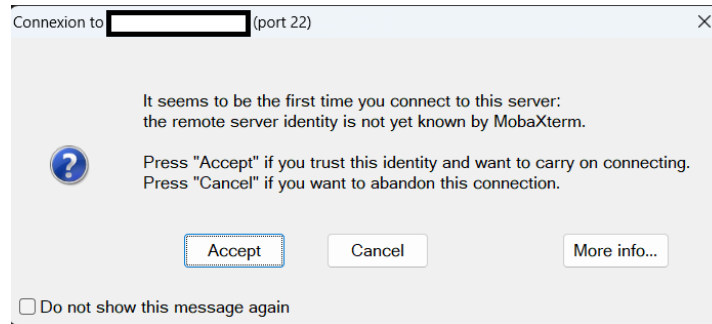


Image Source: Self-created

Connecting to the Linux LPAR (Win)



Important note:

If you see this pop-up, then you are able to access the machine 😊 Click on Accept.

If not, then you are not connected to the same network where our LPAR is located.

You need to use the EduVPN (TUM).

Connecting to the Linux LPAR (mac)

- Open the terminal and type: **ssh** <YOUR LEARN-ID>@<HOSTNAME>
 - For example, if your Learn-ID is learn99 and the hostname is workshop.ucc.cit.tum.de:
`ssh learn99@workshop.ucc.cit.tum.de`
- There should appear a message:
 - *The authenticity of host <HOSTNAME> can't be established*
 - ...
 - *Are you sure you want to continue connecting (yes/no/[fingerprint])?*
- **If you see this message, then you are able to access the machine 😊**
 - Type yes and hit the *Enter* key
 - **If not**, then you are not connected to the same network where our LPAR is located.

You need to use the EduVPN (TUM).

Logging-In

- Use the password from the information sheet we provided for you (*SSH Connection*)
- Note: Type carefully as the terminal will not display any of your password! Use CTRL+u to reset your input to nothing.

Tree-Structured Directories

- Some basics:
 - **ls** command:
 - List the contents of a directory, simply using **ls** lists the contents of the **current** directory
 - ***Try it out now, type ls and hit Enter.***
 - **cd** command:
 - Change the directory, e.g. **cd <name_of_directory>**
 - Use **cd ..** to move up one directory level

Tree-Structured Directories

- **Absolute** path names always begin with the root directory
- **Relative** path names are interpreted starting at the current directory
- Example:
 - Assume our current directory is *hana*
 - Using **ls** would show us the three sub-directories *data*, *log* and *shared*

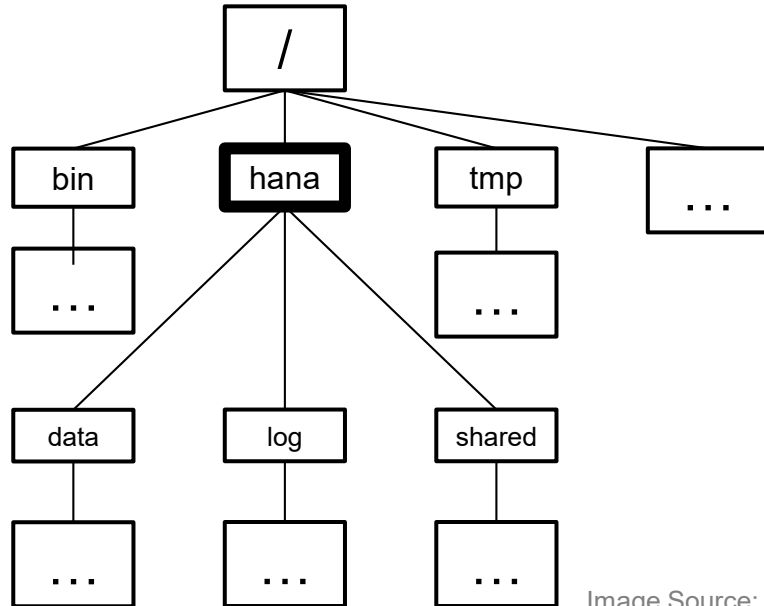


Image Source: Self-created

Tree-Structured Directories

- **Absolute path names** always begin with the root directory.
- **Relative path names** are interpreted starting at the current directory.
- **Example:**

Assume our current directory is *hana*

Using **ls** would show us the three sub-directories *data*, *log* and *shared*

Using **cd log** (relative path name) or
cd /hana/log (absolute path name)
would change the current directory to *log*

Now using **ls** would show the contents
within the directory *log*

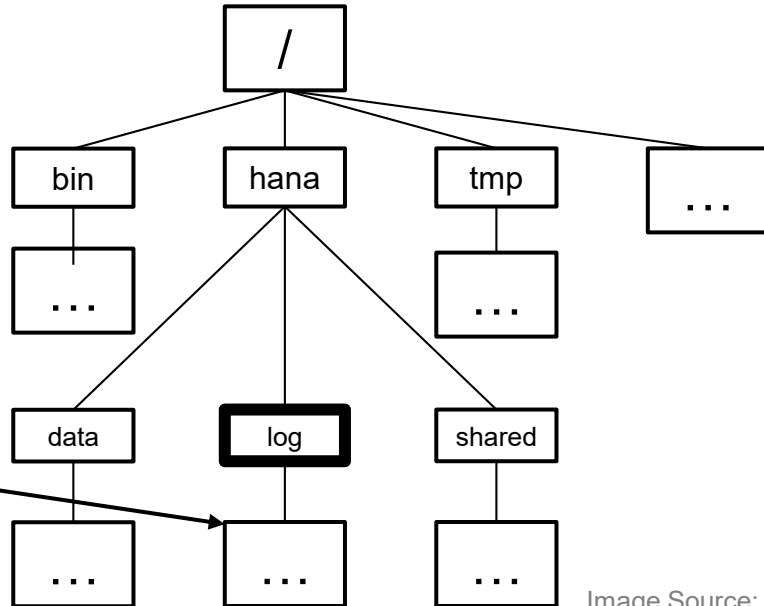


Image Source: Self-created

Tree-Structured Directories

- **Absolute** path names always begin with the root directory
- **Relative** path names are interpreted starting at the current directory

- Example:

- When using
 - `cd ..`
 - `cd ..`

we go up to the root directory /

- We could also do this using the absolute path (`cd /`) or the relative path starting at *log* (`cd ../../`)

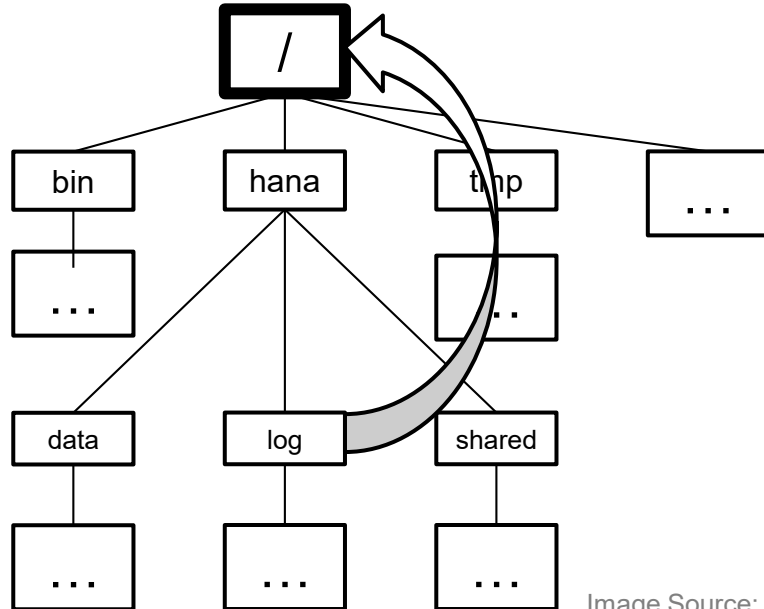


Image Source: Self-created

- If you already changed directory, go back to your home directory (simple `cd`).
- Find the file **hello.txt** (`ls`) and read its contents (`cat`)
- Find the hidden file in your home directory (Also known as "~") and read its contents.
 - Hint: `"tldr ls"`
- Read the contents of **most_important_thing_ever.txt**
 - Hint: `"tldr chmod"`
- In the directory **~/scrambled/**, find the file which contains the text string "youfoundme".
 - Use `cd scrambled` to move into the directory.
 - Hint: `"tldr grep"`

Shell Scripting (Together)

- The `echo` command repeats its input, e.g.:

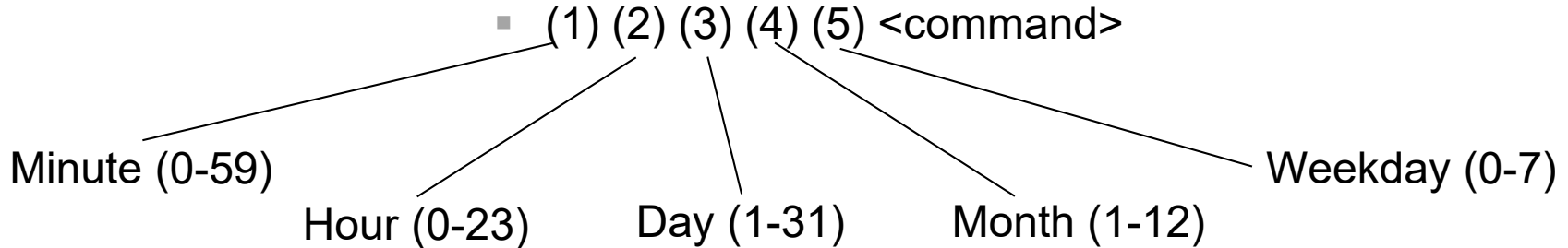
```
> echo "echo"  
echo
```
- ">>" appends the output of a command to a file (it creates the file if it does not yet exist).
 - `echo $(whoami) >> me.txt` creates a file called `me.txt` containing your username.
 - Calling it again adds a new line containing your username to the end of the file.
- Nano is a very basic text editor we can use inside the virtual machine.
 - `nano hello_creator.sh` to create and start editing.

Shell Scripting (Together)

- Inside the text editor:
 - Scripts need to have a `#!/bin/bash` as their first line so the system knows how to interpret them.
 - `echo "hello" >> hello.txt`
 - **CTRL+O to save, then press ENTER, then CTRL+X to exit.**
- **Scripts need to be made executable** using `chmod +x <FILENAME>`
- `./hello_creator.sh` runs the script (assuming it is in your **Personal Working Directory**). The txt file changes everytime the script is executed.

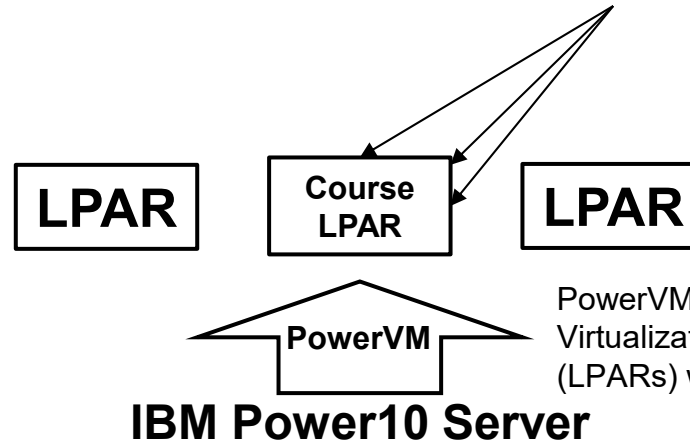
Shell Scripting (Together)

- Crontab is a tool to schedule tasks. You can open it using `crontab -e`
 - `* * * * * /home/<user-name>/shell_scripting/hello_creator.sh > /dev/null`
runs your script every minute into the `/home/<user-name>/hello.txt` file
 - **CTRL+O to save, then press ENTER, then CTRL+X to exit.**



Recap

Student access via different OS users



PowerVM hypervisor for Virtualization, create Logical Partitions (LPARs) with allocated server resources.

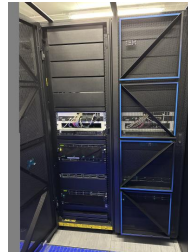


Image Source: Self-created

20-minute COFFEE BREAK

Agenda

- IBM Power Virtualization
- Fundamentals of Red Hat Enterprise Linux (Hands-On!)
- **Podman Containers**
- SAP HANA Deployment (Hands-On!)
- Next Time: IBM Granite / LLM (Hands-On!)

Curriculum Structure

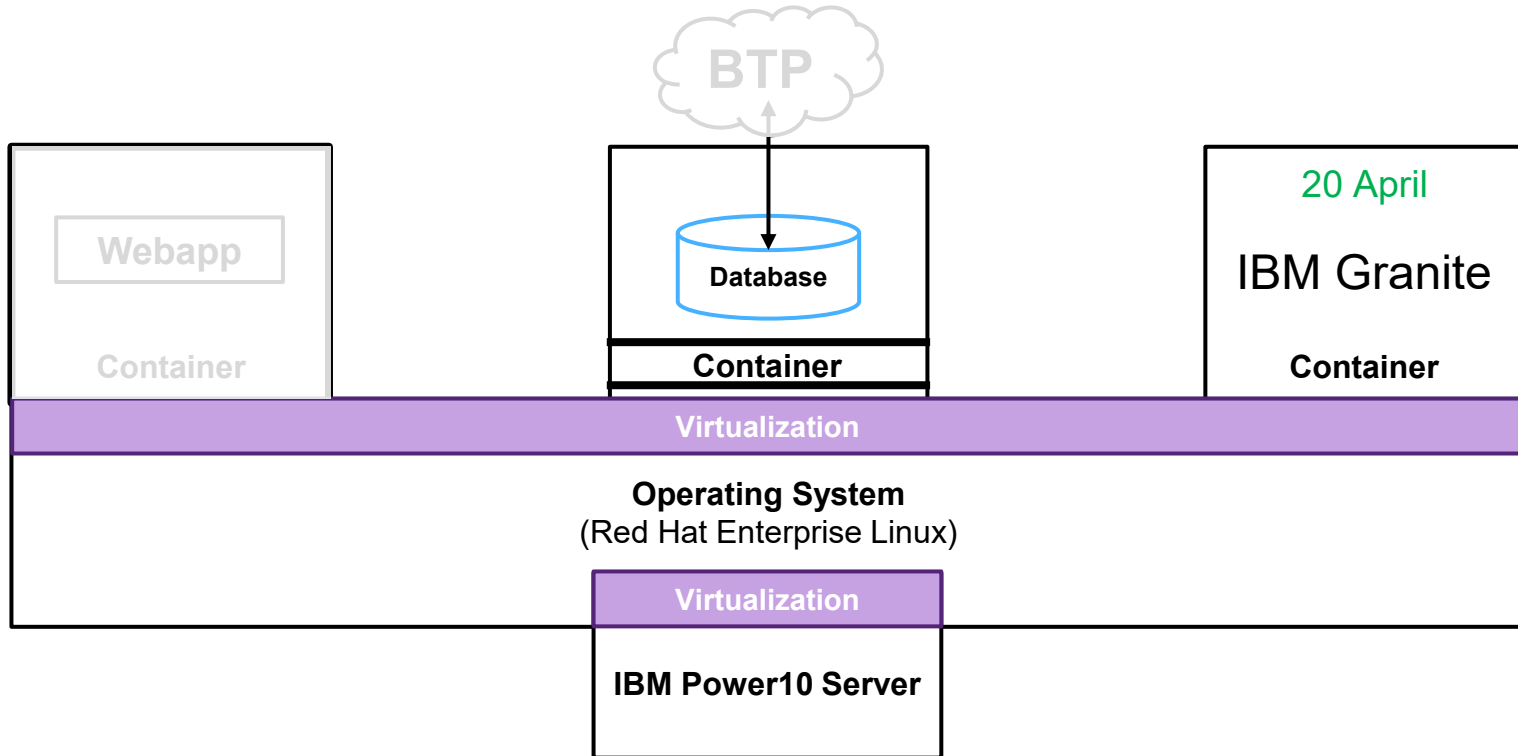
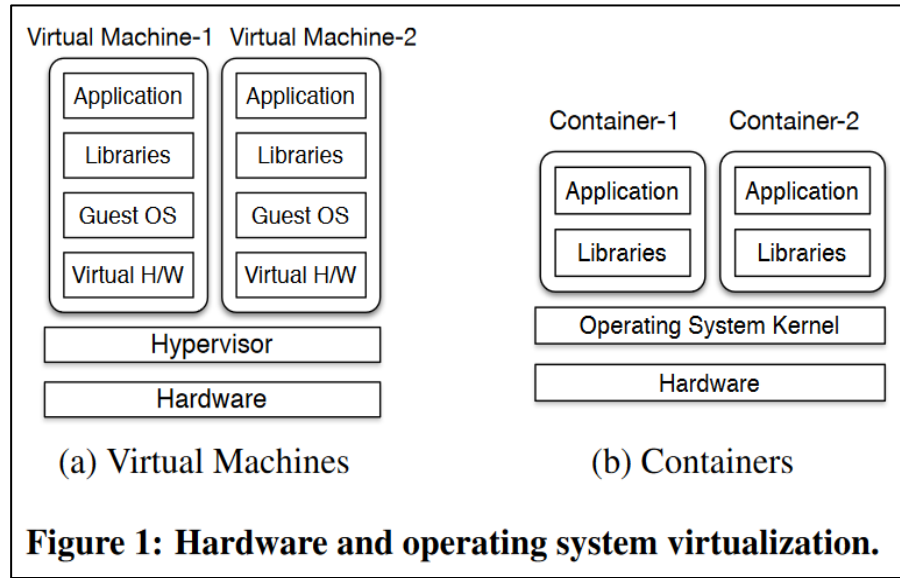


Image Source: Self-created

LPARs & Containers: The Big Picture

- Hardware-Virtualization vs. OS-Virtualization



- **Shared OS kernel**
- **No guest OS**
- **lightweight**

LPARs & Containers: The Big Picture

	Virtual Machine	Docker Containers
Isolation Process Level	Hardware	Operating System
Operating System	Separated	Shared
Boot up time	Long	Short
Resources usage	More	Less
Pre-built Images	Hard to find and manage	Already available for home server
Customised preconfigured images	Hard to build	Easy to build
Size	Bigger because they contain whole OS underneath	Smaller with only docker engines over the host OS
Mobility	Easy to move to a new host OS	Destroyed and recreated instead of moving.
Creation Time	Longer	Within seconds

Image Source: [Potdar et al. \(2019\)](#)

LPARs & Containers: The Big Picture

■ Characteristics of Microservices Architectures:

- **Component size:** Fine-grained
- **Elasticity:** No single point of failure
- **Deployment:** Each service is built and deployed independently
- **Storage mechanism:** Private database
- **Technology:** Heterogeneous
- **Scalability:** Scale on demand

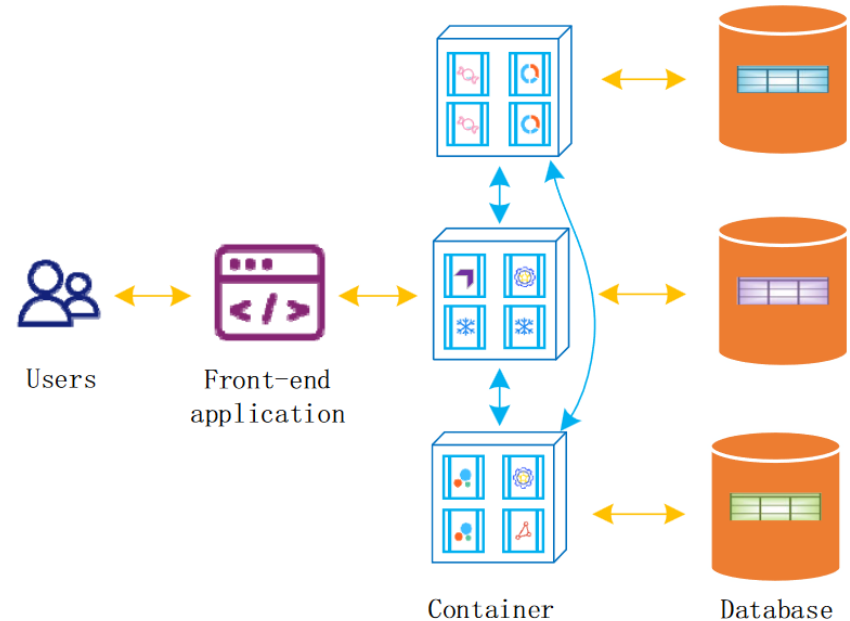
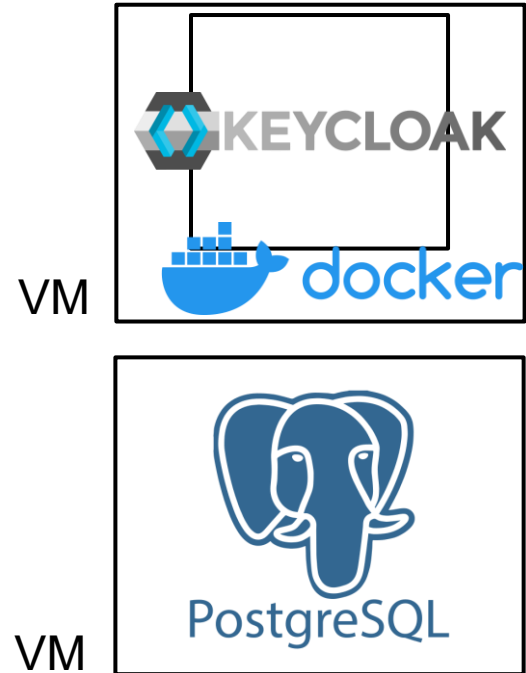


Image Source: [Liu et al. \(2020\)](#)

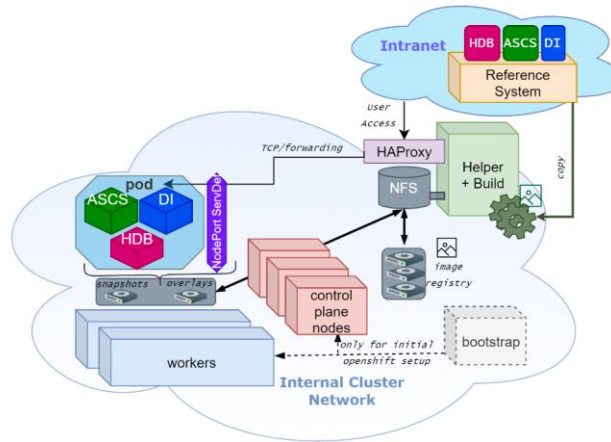
Container Example: Keycloak

- + fast setup (download image, docker run), takes seconds to minutes (e.g. useful for testing scenarios, simulating failures)
- + easy updates (start new container of new image version)



Container Examples: SAP

- Together with IBM, the SAP UCC Munich containerized whole SAP systems.
 - This refers to a S/4HANA ERP system including a full SAP HANA database!
 - Solution architecture:
 - For short-living training systems
 - SAP instances can be built fast (just start the container)



- Key Aspect:
 - Each container only stores the difference to the original HANA database in a snapshot.
=> **Minimal data duplication**, perfect for our use case of demo / training systems!

Container Examples: SAP

- In the next module, each participant will deploy a SAP HANA Express Edition in an own container.
- **Why containers and not give each participant an own LPAR?**
 - Each learner can work in an isolated environment, even though we only use a single LPAR for teaching.
 - Next, if a student makes any mistake, containers allow fast re-deploys to a certain deployment state → *Save Points*
 - Finally, after the course, we simply need to delete the containers. There is no need to create and re-deploy several LPARs for a new course.

Fundamentals: Containerfile

- Information:
 - Text file defining instructions for reproducibly building a container image
 - Includes installation and application configuration steps
- Command-Line Interface:
 - `podman build -t <image-name> .`
 - Builds a new image based on the containerfile in the current directory
 - Podman automatically recognizes containerfiles when they are named *Dockerfile* or *Containerfile*

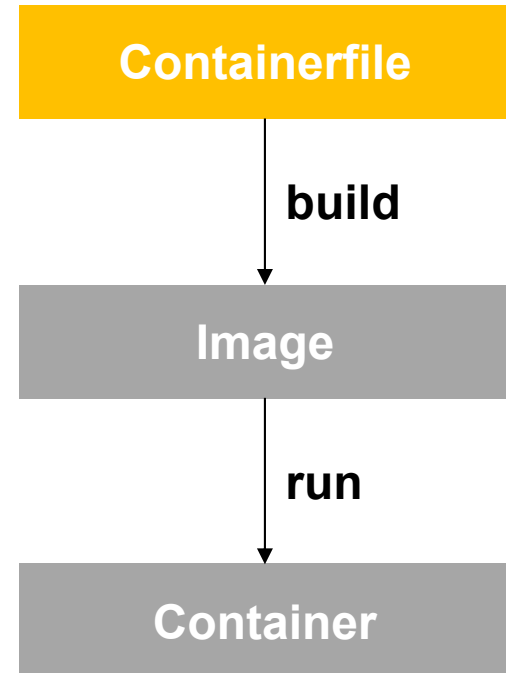


Image Source: Self-created

Fundamentals: Image

- Information:
 - Foundation for creating and running containers
 - Snapshot of an application and its dependencies, packaged in a portable format
- Command-Line Interface:
 - `podman images`
 - Lists available container images
 - `podman run -it <image-name> /bin/bash`
 - Creates a new container based on a specified image and starts an interactive shell session

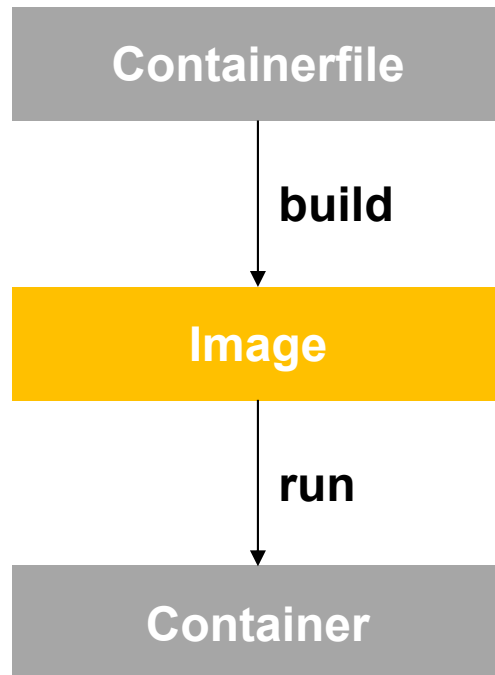


Image Source: Self-created

Fundamentals: Container

- Information:
 - Lightweight environment encapsulating an application along with its dependencies
- Command-Line Interface:
 - `podman ps -a`
 - Lists all running and stopped containers
 - `podman start (stop) <container-id>`
 - Starts (stops) an existing container
 - `podman exec -it <container-id> /bin/bash`
 - Executes an interactive shell session for a container
 - `podman commit <container-id> <image-name>`
 - Creates a new image based on the current container state
 - `podman rm <container-id>`
 - Removes a stopped container

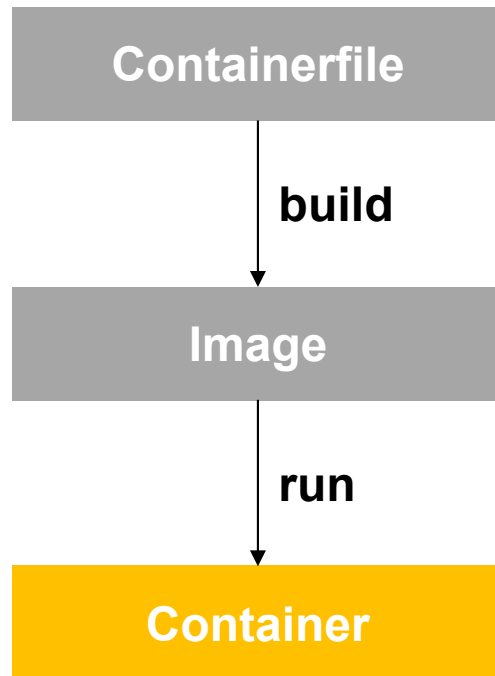


Image Source: Self-created

Agenda

- IBM Power Virtualization
- Fundamentals of Red Hat Enterprise Linux (Hands-On!)
- Podman Containers
- **SAP HANA Deployment (Hands-On!)**
- Next Time: IBM Granite / LLM (Hands-On!)

Curriculum Structure

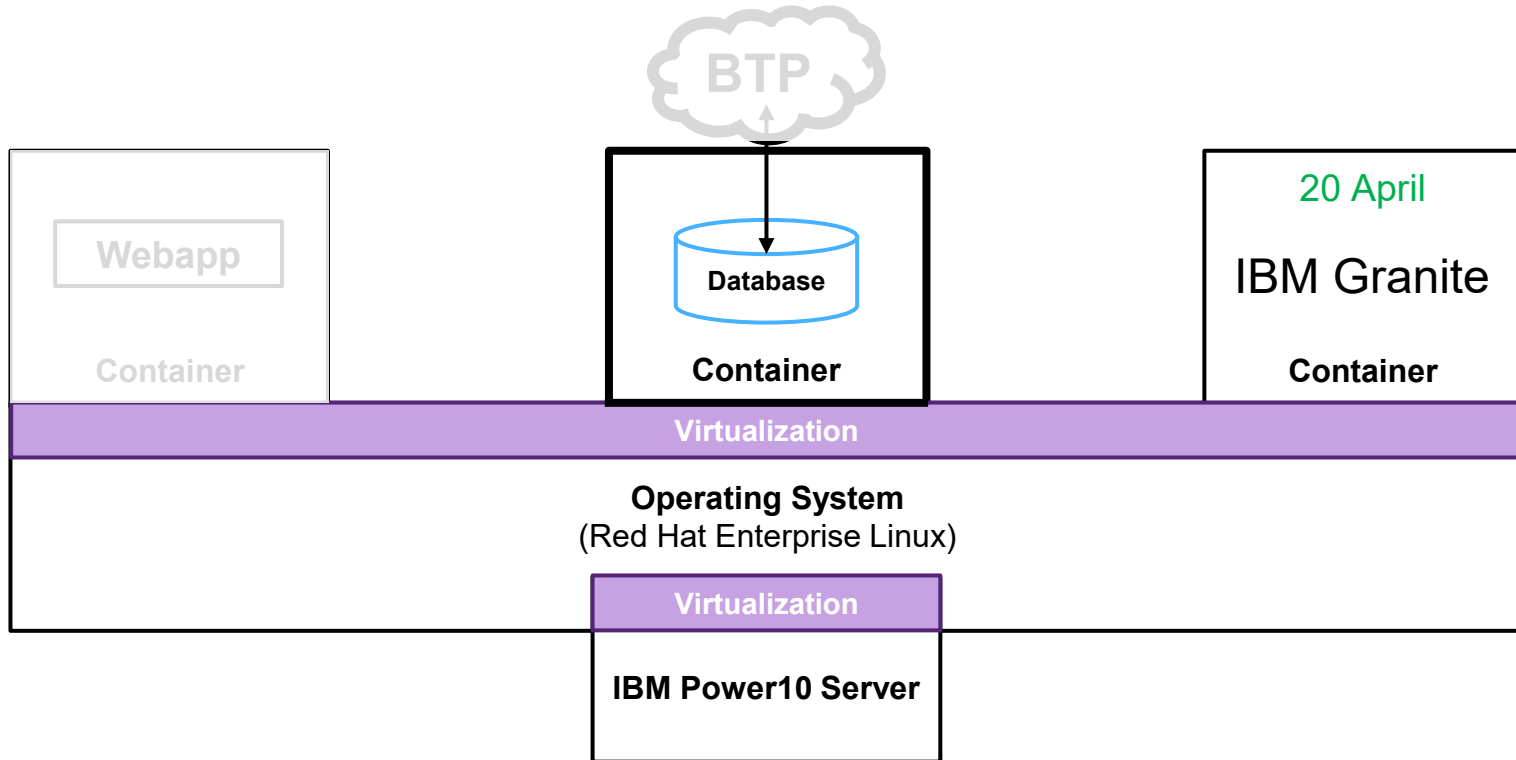


Image Source: Self-created

SAP HANA: The Big Picture (Recap)

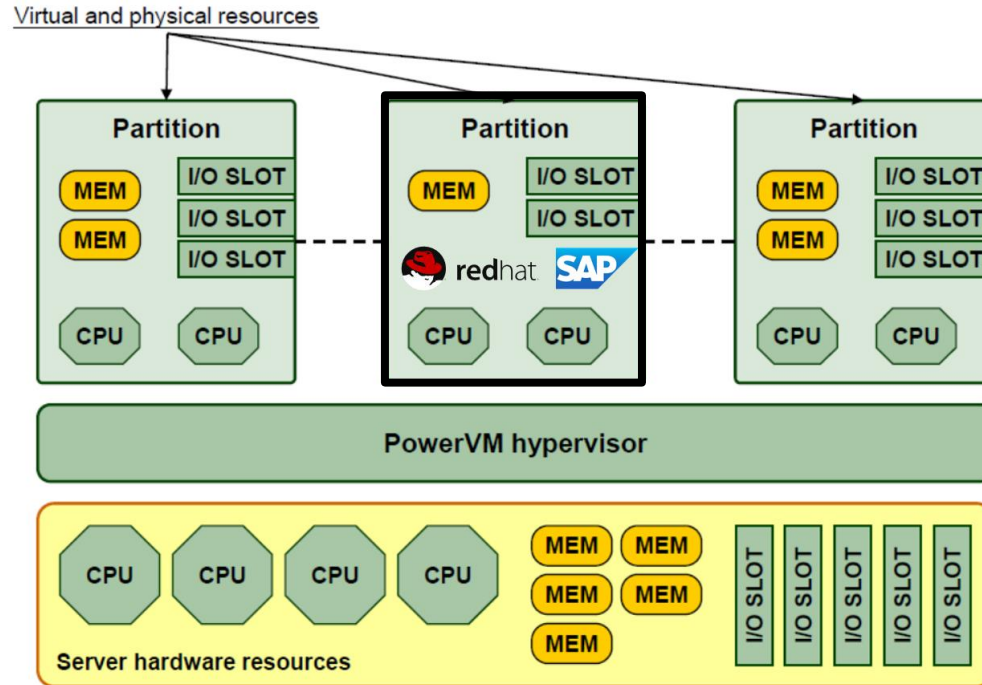


Image Sources:

[Introduction to IBM PowerVM](https://de.wikipedia.org/wiki/Datei:Red_Hat_logo.svg)

https://de.wikipedia.org/wiki/Datei:Red_Hat_logo.svg

https://de.wikipedia.org/wiki/SAP#/media/Datei:SAP_2011_logo.svg

SAP HANA: The Big Picture (Recap)

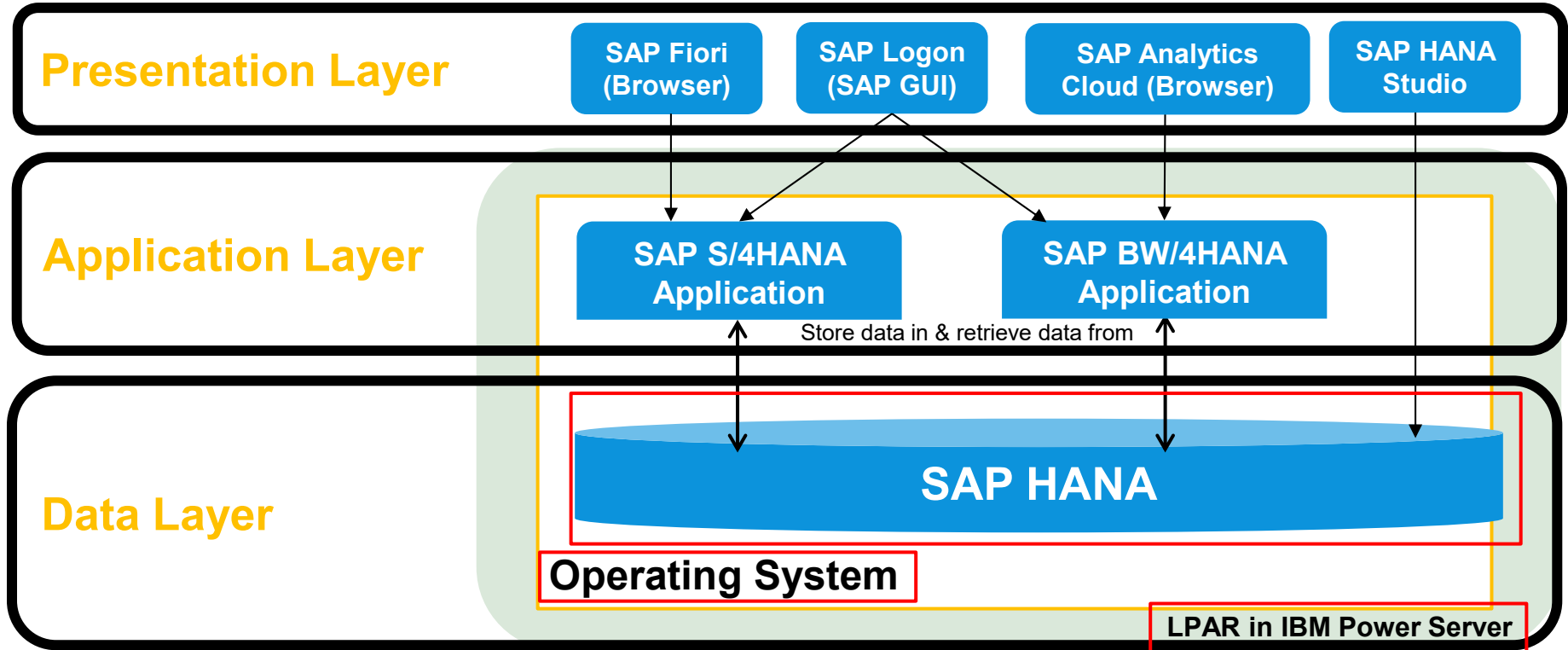
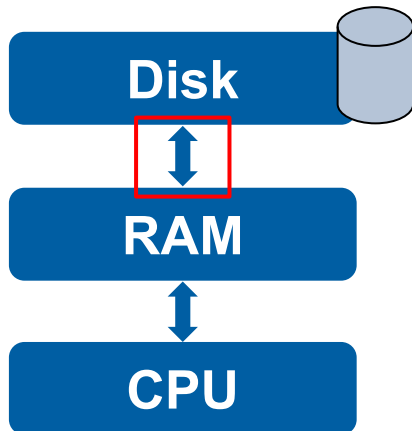


Image Source: Self-created

SAP HANA: An In-Memory Database

- Without high-capacity RAM
 - Database stored on disk
 - **Bottleneck:** Latency between disk and RAM



- With high-capacity RAM (SAP HANA)
 - Data stored in-memory
 - Bottleneck: Latency between CPU and RAM
 - Huge response time improvement

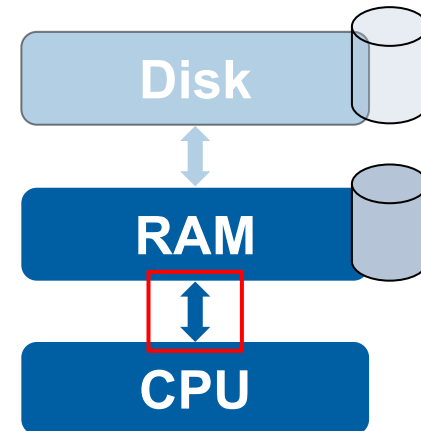


Image Source: Self-created

SAP HANA: An In-Memory Database

■ Hot Data

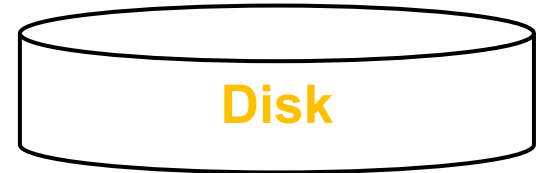
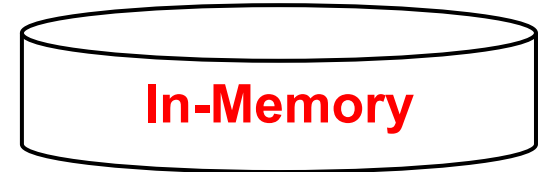
- Business-critical data
- For Real-Time Processing and Analysis

■ Warm Data

- Data with low SLA in terms of performance
- Less frequent access
- Cost-effective storage layer

■ Cold Data

- Very limited Access
- Cost-effective storage layer, separately managed from SAP HANA database



SAP HANA: An In-Memory Database

- SAP HANA databases utilize Structured Query Language (SQL), incorporating a distinct SQL dialect as outlined in the [reference guide](#).
- Differences between SQL dialects are small, e.g. in SAP HANA SQL you will have to use *CAST* or *TO_VARCHAR* for data type conversion and not *CONVERT* (MSSQL).
- However, these small changes make Query Migrations between different databases complex!

SAP HANA: An In-Memory Database

- SQL Query: **All columns**

```
SELECT *  
FROM PUBLIC_SCHEMA.PRODUCT  
WHERE  
CATEGORY = 'Electronics'  
and  
PRICE > 100
```

Table:

	PRODUCT_ID	PRODUCTNAME	INSTOCK	PRICE	CATEGORY	RELEASEDATE
1	1	Laptop	20	899,99	Electronics	01.03.2023
2	2	Sneakers	50	79,99	Fashion	15.02.2022
3	3	Coffee Maker	15	129,95	Home A...	10.04.2020
4	4	Backpack	30	49,99	Outdoor ...	20.03.2021
5	5	Smartwatch	35	149,99	Electronics	12.05.2023
6	6	Wireless Earbu...	25	59,99	Electronics	08.06.2022
7	7	Yoga Mat	40	29,99	Fitness	25.04.2022
8	8	Blender	18	89,95	Kitchen A...	26.02.2024

Results:

	PRODUCT_ID	PRODUCTNAME	INSTOCK	PRICE	CATEGORY	RELEASEDATE
1	1	Laptop	20	899,99	Electronics	01.03.2023
2	5	Smartwatch	35	149,99	Electronics	12.05.2023

Image Source: Self-created

SAP HANA Installation

SAP HANA Installation: Background



- SAP HANA Express:
 - Instead of installing SAP HANA, we install the EXPRESS edition which is a free version for prototyping use cases ([Docs](#)).
 - SAP HANA Express is resource-constrained **up** to 32GB of RAM.
 - Suitable for our case as every course participant shall install their own database.

- Container as Host:
 - You will install the database on a container and not on the LPAR itself.
 - This lightweight approach is ideal for our educational use case, in which databases are not intended to be permanent, but need to be deployed and deprovisioned efficiently.
 - Also, it allows us that each participant operates in their own isolated environment.

SAP HANA Installation: Build Image

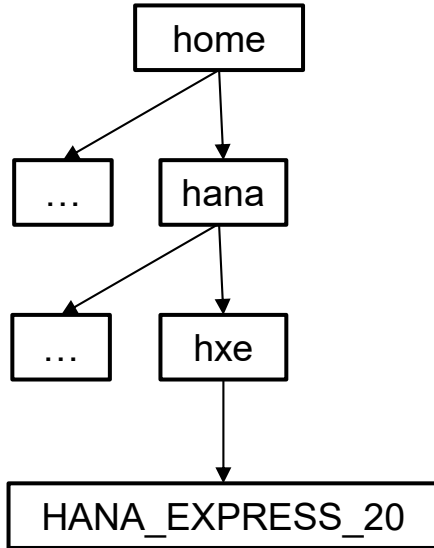
- From your home directory (use **cd ~** to get there), change to the *HANA* directory.
- Inspect the *Dockerfile* (**cat**).
- The image on the right shows the most interesting parts of the Dockerfile.
- What are these commands doing?

```
# COPY contents from machine to image
COPY /hxe/ /jail

# Install packages necessary for SAP HANA Express installation
RUN yum install -y libxcrypt-compat
RUN yum install -y chkconfig
RUN yum install -y iproute
RUN yum install -y libaio
RUN yum install -y libatomic
RUN yum install -y numactl-libs
RUN yum install -y libtool-ltdl
RUN yum install -y procps-ng
RUN yum install -y ncurses
RUN yum install -y hostname

# Adjust Access within image
RUN chmod +x /jail/setup_hxe.sh
RUN chmod -R +x /jail/HANA EXPRESS 20/*
```

SAP HANA Installation: Build Image



HANA_EXPRESS_20:

Directory which contains data for SAP HANA Express installation

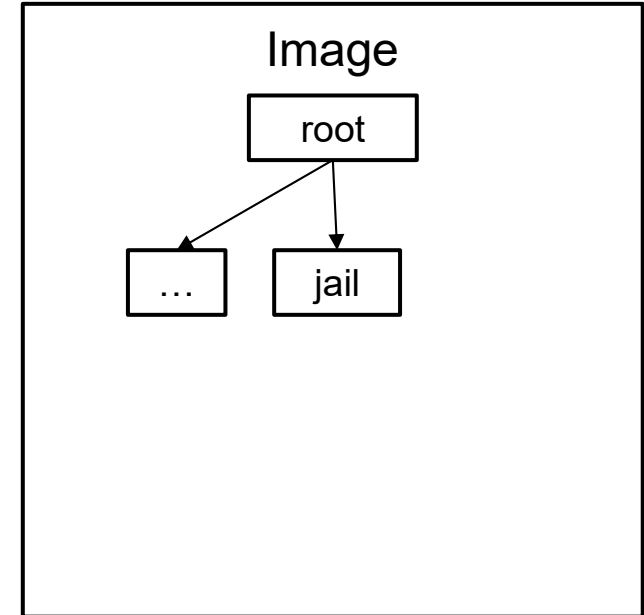
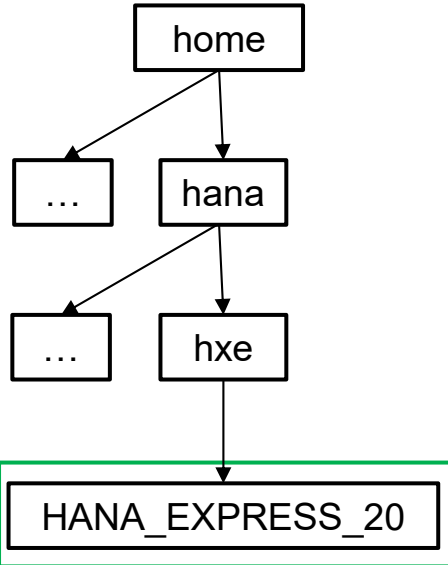
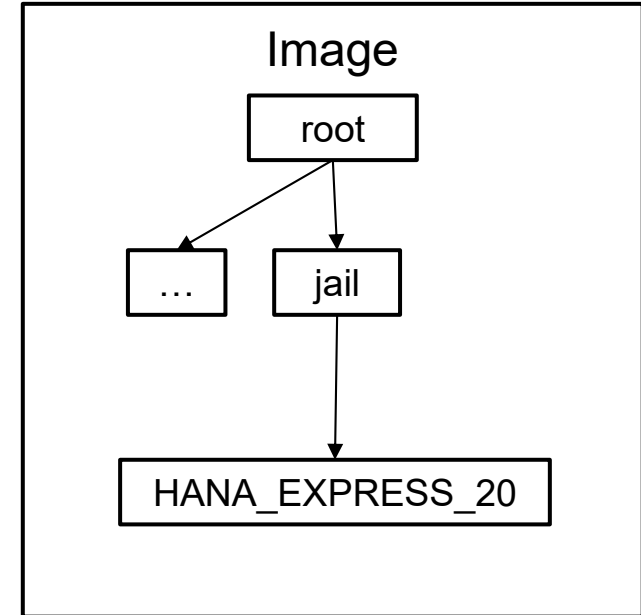


Image Source: Self-created

SAP HANA Installation: Build Image



COPY /hxe/ /jail

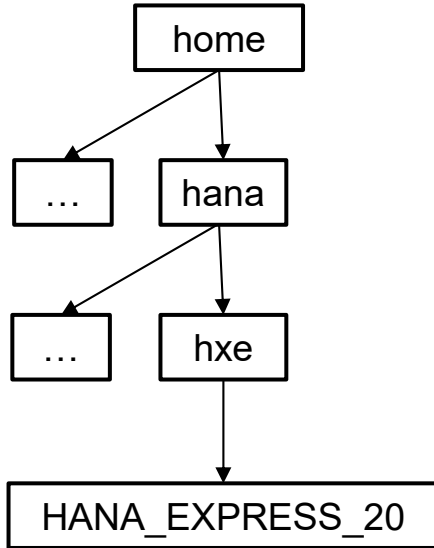


HANA_EXPRESS_20:

Directory which contains data for SAP HANA Express installation

Image Source: Self-created

SAP HANA Installation: Build Image



RUN yum install -y <lib1>
RUN yum install -y <lib2>
...

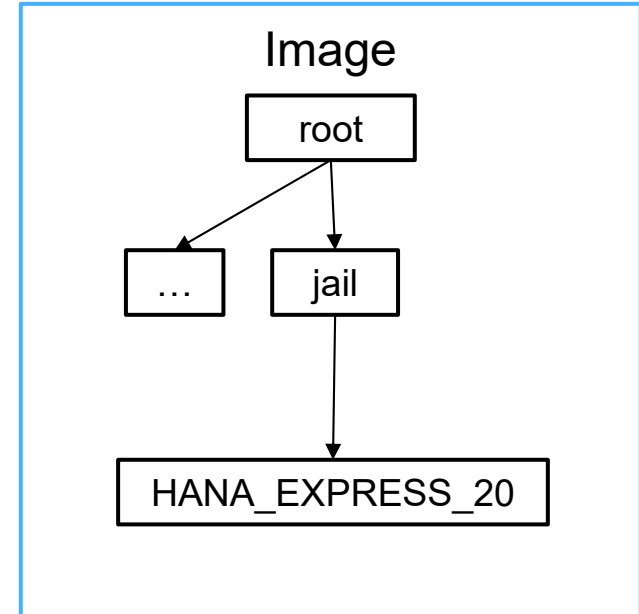
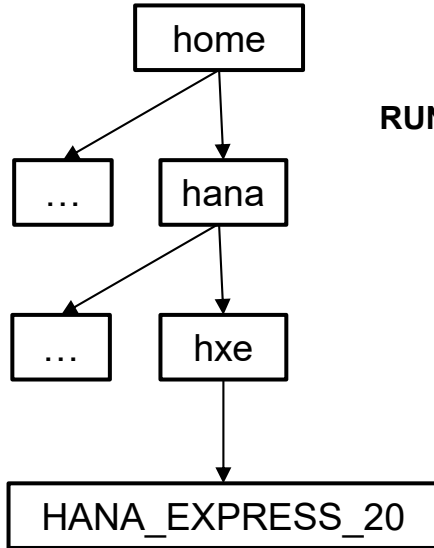


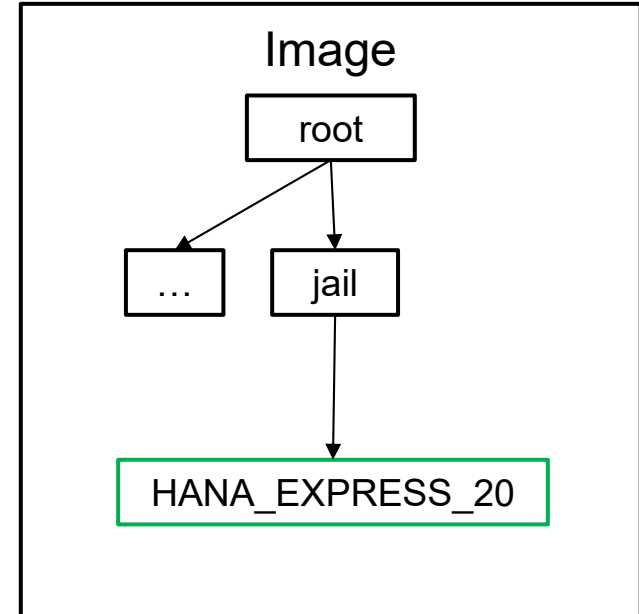
Image is now „**SAP HANA Express**“-ready.
Installation will not throw any errors „lib X is missing“.

Image Source: Self-created

SAP HANA Installation: Build Image



RUN `chmod -R +x /jail/HANA_EXPRESS_20/*`



All files in *HANA_EXPRESS_20* directory are executable
(no permission errors will occur).

Image Source: Self-created

SAP HANA Installation: Build Image

- Now, based on this Dockerfile, build a new image called *redhat4hana*.
- Tip: The previous chapter (see **Fundamentals: Containerfile**) shows you how to build images.
- Verify that you successfully built the image by using the command **podman images**

```
[lecturer@RZCurrTest2 HANA]$ podman images
```

REPOSITORY	TAG	IMAGE ID	CREATED	SIZE
localhost/redhat4hana	latest	143abb7844c5	3 weeks ago	5.46 GB
registry.access.redhat.com/ubi9/ubi	9.2-755.1697625012	9809f8a11efd	14 months ago	263 MB

SAP HANA Installation: Start Container

- You still need to operate within the *HANA* directory (use **pwd** to check).
- Use nano to open and edit the file *podman_run.sh*
 - Adjust the *System Port* and the *Database Port* (**check information sheet for finding the ports assigned to you! 39913 and 39915 are just examples**).
 - The image below shows an example of a learner who was assigned the instance number 99.
 - Use **CTRL+O** to save -> **ENTER** -> **CTRL+X** to exit.

```
GNU nano 5.6.1 podman_run.sh
podman run -i -d --name lecturer_HANA2 -e "container=docker" --ipc=host -t -h saphost1 -p 39913:39913 -p 39915:39915 redhat4hana2
```

Changed from **-p <Port SYSDB>:<Port SYSDB> -p <Port TENDB>:<Port TENDB> to
-p 39913:39913 -p 39915:39915**

Image Source: Self-created

SAP HANA Installation: Start Container

- Next, run the shell script via *podman_run.sh*
- Tip: In the Linux chapter (see **Shell Scripting (Together)**), we showed how to run shell scripts.
- Before running the script, you need to make sure that your script is executable
 - `chmod +x <FILENAME>`
- After running the script, use `podman ps` to check that the container now exists!

SAP HANA Installation: Start HDBLCM

- By running the container, we created an isolated environment, which is „SAP-ready“.
- You do not need to install any dependencies, but can directly install the SAP HANA Express database.

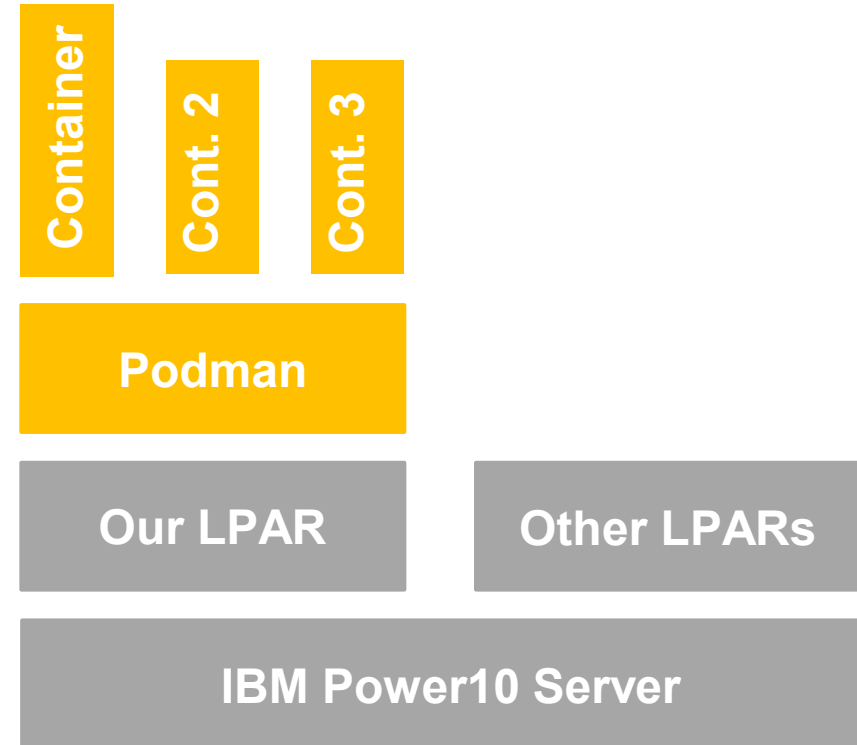


Image Source: Self-created

SAP HANA Installation: Start HDBLCM

- Use *podman exec -it <container ID> /bin/bash* to operate within the container. Or *podman exec -it <container-name> /bin/bash*
- If you did not change the name, it is: `podman exec -it HANA /bin/bash`
- You can check the name via `podman ps`
- (use *exit* if you want to go „back“ and change to the learn / OS user on the LPAR)

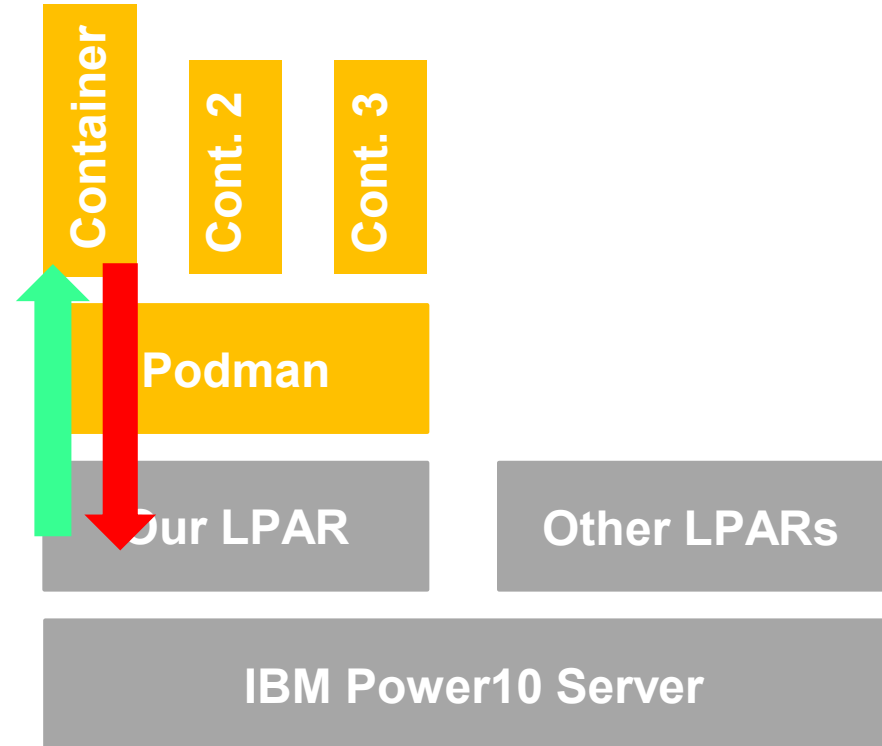


Image Source: Self-created

SAP HANA Installation: Start HDBLCM

- Next, change to the `/jail/HANA_EXPRESS_20/DATA_UNITS/HDB_LCM_LINUX_PPC64LE` directory.
- You can use the **Tab** key to auto-complete your inputs.

```
@localhost:/jail/HANA_EXPRESS_20/DATA_UNITS/HDB_LCM_LINUX_PPC64LE
[root@saphost1 /]# ls
afs boot etc jail lib64 media opt root sbin sys usr
bin dev home lib lost+found mnt proc run srv tmp var
[root@saphost1 /]# cd /jail/HANA_EXPRESS_20/DATA_UNITS/HDB_LCM_LINUX_PPC64LE/
[root@saphost1 HDB_LCM_LINUX_PPC64LE]# ls
SIGNATURE.SMF descriptors.d hdblcmm instruntime
adapters.d filelist.hdblcmm_remote_check hdblcmmgui operations.d
configurations filelist.resident hdblcmmweb resources
[root@saphost1 HDB_LCM_LINUX_PPC64LE]# ./hdblcmm

SAP HANA Lifecycle Management - SAP HANA Express Edition 2.00.072.00.1690304772
*****

Choose an action

Index | Action | Description
-----|-----|-----
1 | install | Install new system
2 | Exit (do nothing) |

Enter selected action index [2]:
```

SAP HANA Installation: Start HDBLCM

- Run the *Hana Database Lifecycle Manager* using the command **./hdblcmm**

```
@localhost:/jail/HANA_EXPRESS_20/DATA_UNITS/HDB_LCM_LINUX_PPC64LE
[root@saphost1 /]# ls
afs boot etc jail lib64 media opt root sbin sys usr
bin dev home lib lost+found mnt proc run srv tmp var
[root@saphost1 /]# cd /jail/HANA_EXPRESS_20/DATA_UNITS/HDB_LCM_LINUX_PPC64LE/
[root@saphost1 HDB_LCM_LINUX_PPC64LE]# ls
SIGNATURE.SMF descriptors.d hdblcmm instruntime
adapters.d filelist.hdblcmm_remote_check hdblcmmgui operations.d
configurations filelist.resident hdblcmmweb resources
[root@saphost1 HDB_LCM_LINUX_PPC64LE]# ./hdblcmm

SAP HANA Lifecycle Management - SAP HANA Express Edition 2.00.072.00.1690304772
*****

Choose an action

Index | Action | Description
-----|-----|-----
1 | install | Install new system
2 | Exit (do nothing) |

Enter selected action index [2]:
```

SAP HANA Installation: Hands-On Task

■ Independent Task:

Goal:

Install a SAP HANA Express database.

Pressing Enter
without any inputs
will take over the
suggestions in the
square brackets.

You need to **install a
new system**, but
**only a SAP HANA
server**, no additional
components!

Use the info sheet to
find the **database
password & the
instance number**
assigned to you.

Please leave everything else as
it is! (Just press Enter)
Installation path is
/hana/shared, local host name
is **saphost1**, System ID is
HXE, User Group ID is **79**.

• A successful installation ends with:

**If you think you did a mistake while entering your password,
press CTRL+u to reset your password inputs.**

```
Creating Component List...
SAP HANA Express Edition System installed
Log file written to '/var/tmp/hdb_HXE_hdblcm_install_2023-12-17_15.34.44/hdblcm.log' on host 'saphost1'.
saphost1:/jail/HANA_EXPRESS_20/DATA_UNITS/HDB_LCM_LINUX_PPC64LE # █
```

Image Source: Self-created

SAP HANA Installation: Start HDBLCM

- Use **exit** to go back to the LPAR level.
- If you want to, you can log-in to the Container as the database admin:
 - Therefore, use `podman exec -u hxeadm -it <container name> /bin/bash`
 - Important DB Admin commands:
 - `HDB info`
 - Note: If `HDB info` does not show the hdb services, you can try a restart
 - `HDB stop` (stops the database!)
 - `HDB start`

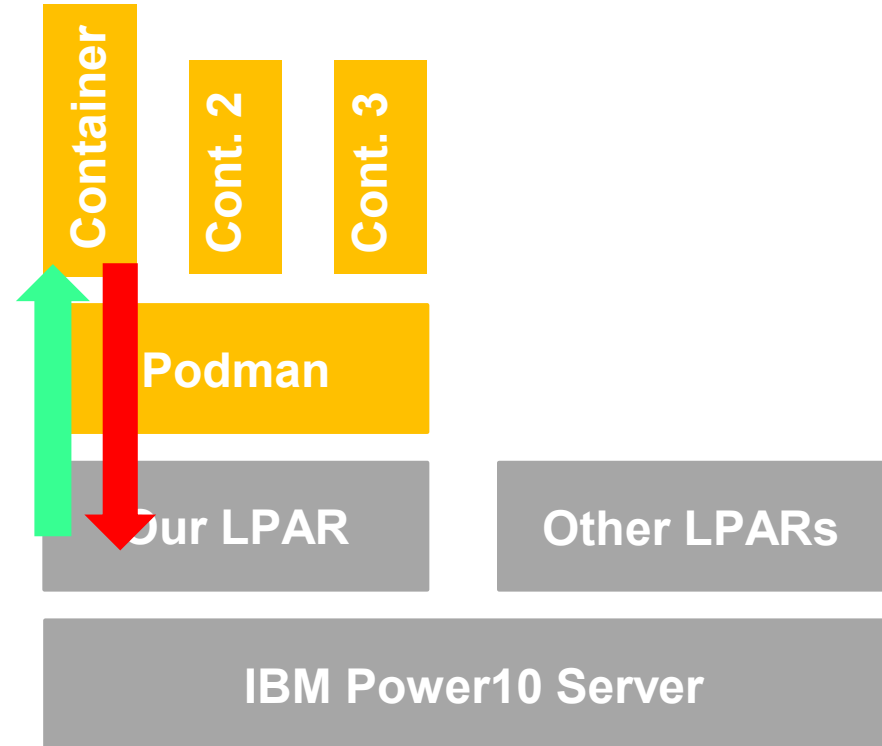
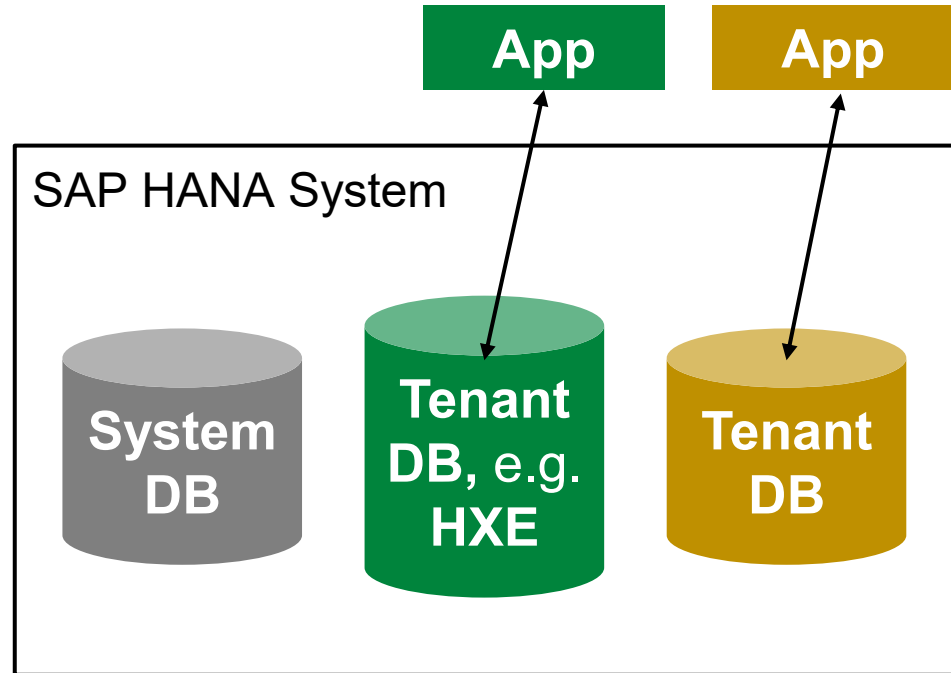


Image Source: Self-created

SAP HANA Insights

- SAP HANA database systems consist of 1 system database and 1-many tenant databases.
- The system database is the central management and administration database, responsible for handling tasks such as user authentication, authorization and lifecycle-management of tenant databases.
- Tenant databases are used to store and process application data.



Source: Self-created

How to fill in data?

- SAP Tools
 - SAP HANA Studio (old)
 - SAP HANA Cockpit (new)
- Or: Terminal

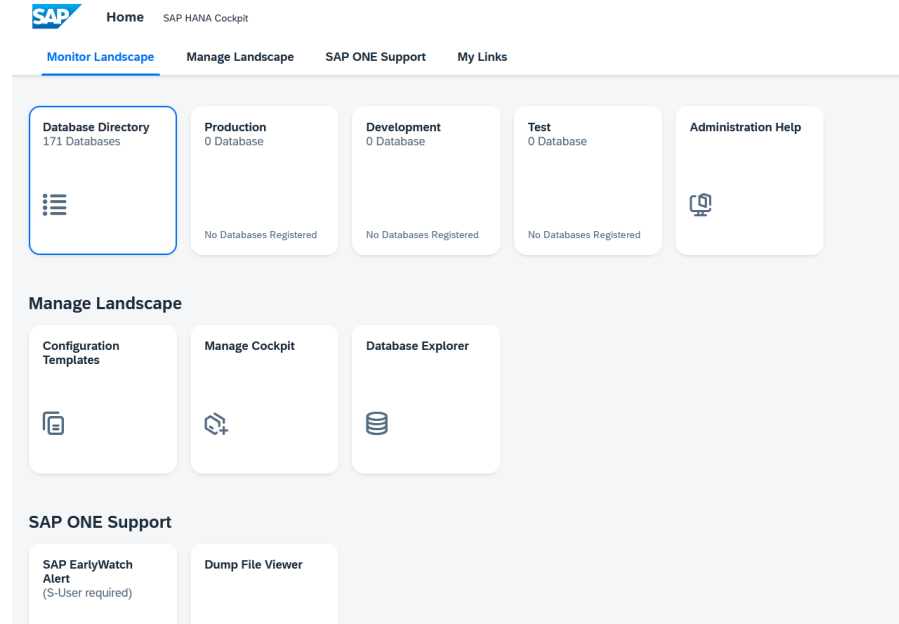


Image Source: Self-created

How to fill in data?

- If learnX@sose26dc.ucc.cit.tum.de
 - Get into container as hxeadm
 - *podman exec -u hxeadm -it <container name> /bin/bash*

- If root@saphost1
 - Get out of container, and again into container as hxeadm
 - exit
 - *podman exec -u hxeadm -it <container name> /bin/bash*

- If hxeadm@saphot1
 - Correct.

How to fill in data?

- Change to SQL console.
 - `hdbsql -i <Instance Number> -u SYSTEM`
 - E.g. `hdbsql -i 99 -u SYSTEM` if your instance number is 99
 - Then, enter the password you chose during the HANA installation, e.g. Password1
 - Then:

```
Welcome to the SAP HANA Database interactive terminal.  
  
Type:  \h for help with commands  
       \q to quit  
  
hdbsql HXE=>
```

Example to create a small SQL environment:

- **CREATE SCHEMA** PUBLIC_SCHEMA;
- **GRANT SELECT ON SCHEMA** PUBLIC_SCHEMA **TO PUBLIC**;
- **CREATE TABLE** PUBLIC_SCHEMA.Products (Product_ID **VARCHAR**(255) **PRIMARY KEY**, Product **VARCHAR**(255), Price **FLOAT**);
- **CREATE TABLE** PUBLIC_SCHEMA.Sales (Sales_ID **INT PRIMARY KEY**, Product_ID **VARCHAR**(255), Day **DATE**, Time **TIME**, **FOREIGN KEY** (Product_ID) **REFERENCES** PUBLIC_SCHEMA.Products(Product_ID));
- **CREATE VIEW** PUBLIC_SCHEMA.ProductSales **AS SELECT** p.Product, p.Price, s.Day, s.Time **FROM** PUBLIC_SCHEMA.Sales s **LEFT JOIN** PUBLIC_SCHEMA.Products p **ON** p.Product_ID = s.Product_ID;

Agenda

- IBM Power Virtualization
- Fundamentals of Red Hat Enterprise Linux (Hands-On!)
- Podman Containers (Hands-On!)
- SAP HANA Deployment (Hands-On!)
- **Next Time: IBM Granite / LLM (Hands-On!)**

Agenda

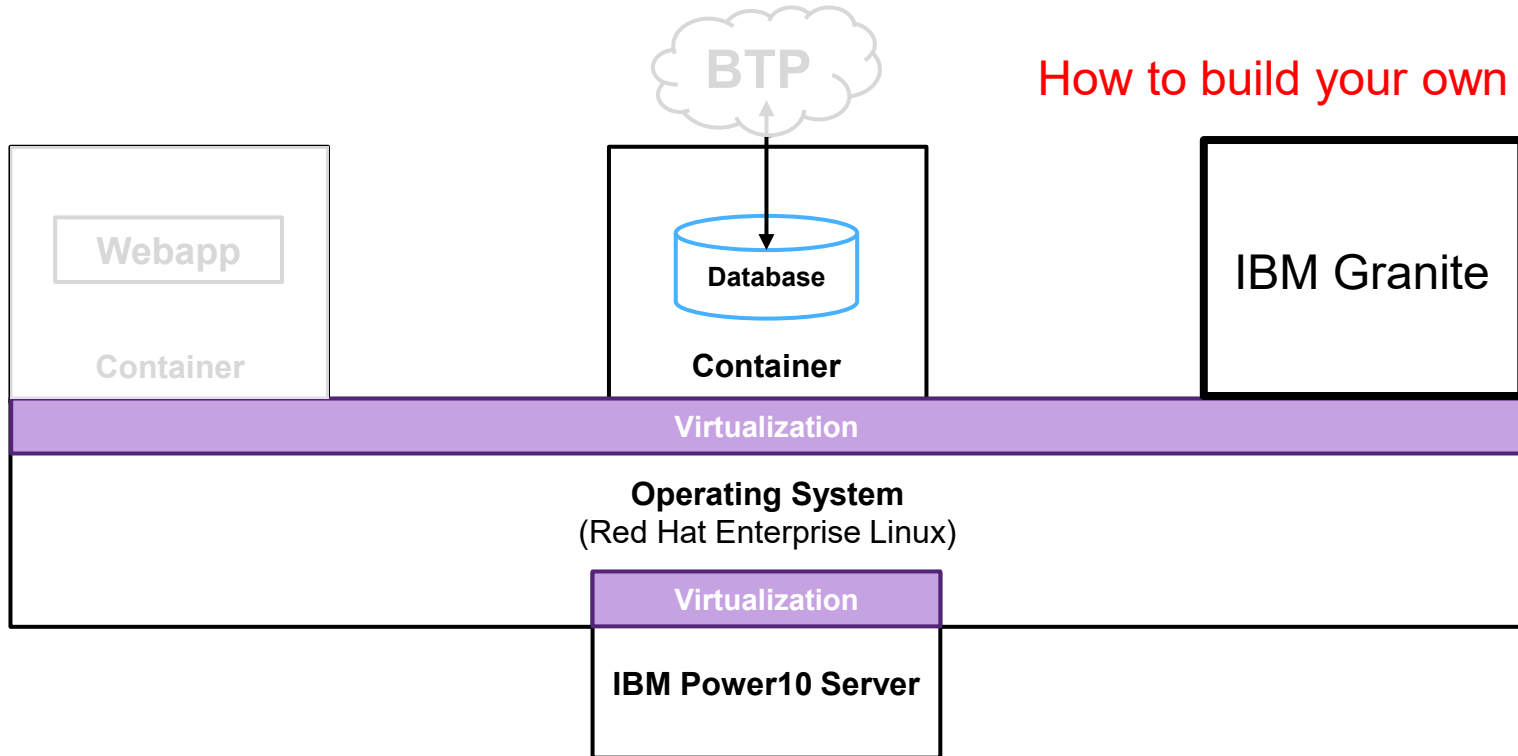


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Appendix

Contact information:

Please message max.niedermeier@tum.de if you have any questions regarding this curriculum.